23L-0588 — VisionGuard

AI-generated versus real-image classification service

Student: Muhammad Abdul Raheem

Project: 23L-0588_A2_Ops

GitHub: github.com/abdul-raheem-fast/23L-0588_A2_Ops

DagsHub: dagshub.com/abdul-raheem-fast/23L-0588_A2_Ops

Live URL: 23l-0588-a2-ops.ghauri.prod.page

Final repository commit: e923d98

This report documents data understanding, model development, experiment tracking, Flask serving, database persistence, containerization, deployment, and real execution evidence for VisionGuard.

Disclosure

An AI assistant was used to help structure the implementation, identify reproducibility requirements, and review wording. All commands, runs, screenshots, account actions, metrics, and interpretations in this report must be produced and verified by the student.

1. Data understanding and versioning

1.1 Environment

The workflow was executed in an isolated Colab runtime and the environment was checked before training. Python, TensorFlow, Keras, Pillow, MLflow, and Docker versions were recorded in the run evidence.

The Colab runtime selected a T4 GPU. TensorFlow 2.18.0 was unavailable for that runtime, so the run used TensorFlow 2.21.0; this version change is recorded rather than hidden.

1.2 Dataset audit

The audit script records the class label, file size, dimensions, colour mode, file format, SHA-256 content hash, readability, and duplicate status for every file. This makes data-quality problems visible before the training split is created.

The Colab run audited 975 images: 539 from AiArtData and 436 from RealArt. No files were unreadable, and two exact duplicate files were identified by matching content hashes. The class imbalance is modest, so stratified splits were used to preserve class proportions.

1.3 Visual EDA

The audited metadata was used to generate dataset_eda.svg. The sample grid shows representative examples from both classes. The width, height, and file-size distributions show substantial variation, including very large source images, which supports resizing and batched loading in the input pipeline. These distributions describe the collected files rather than image semantics and are not treated as proof of a class-specific visual feature.

1.4 DVC

The repository includes the DVC pipeline configuration in dvc.yaml and params.yaml. The pipeline separates dataset auditing from split generation and records the seed and image size in configuration. A separate authenticated DVC push/pull screenshot was not captured, so remote transfer is not claimed as completed evidence.

2. Model development

The tf.data pipeline reads the reproducible split CSVs, decodes each image, converts it to RGB, resizes it to 224×224, batches samples, and prefetches batches for efficient execution. Training-only augmentation is applied before batching; validation and test images are only resized and normalized. Resizing uses a fixed square target, which provides a consistent tensor shape but can change the original aspect ratio.

The baseline CNN uses convolution, pooling, normalization, and dropout layers followed by a sigmoid output for binary classification. MobileNetV2 was then loaded without its classifier, trained first with the backbone frozen, and fine-tuned by unfreezing the final block at a lower learning rate. Early stopping, checkpointing on validation loss, and learning-rate reduction were used to retain the best generalizing checkpoint.

The executed Colab run selected the fine-tuned MobileNetV2 model. Its recorded test metrics are in champion_metrics.json: accuracy 0.7449, precision 0.8718, recall 0.6296, F1 0.7312, and ROC-AUC 0.8695. The precision/recall difference indicates that the selected threshold favours fewer false AI-generated predictions at the cost of missed positives. For this product, incorrectly calling a real photograph AI-generated is the more damaging user-facing error, so the precision-oriented operating point is preferable to maximizing recall alone. The evidence notebook generated confusion_matrix.png, roc_curve.png, error_analysis_grid.png, and error_analysis.csv.

3. MLflow and DagsHub

MLflow tracking was configured against DagsHub. Each executed experiment recorded the model family, training configuration, validation history, and final evaluation metrics, allowing the baseline and MobileNetV2 approaches to be compared from the run pages. The retained screenshots show the tracking configuration, completed runs, and final training output. Independent model-registry version and alias evidence was not captured.

4. Flask application

The model is loaded once during application startup. Upload validation checks the extension, size, and decoded image without persisting the upload. The result shows the label, confidence, and caveat; /history reads the request trail and /health reports service state.

5. Database trail

Each successful prediction stores a UTC timestamp, sanitized filename, decision, confidence, model name, and model version. Uploaded image contents are not persisted. SQLite is used by the free deployment, while the application also supports PostgreSQL through environment-based configuration.

6. Docker and Compose

The Docker image installs serving dependencies separately from training and MLflow tools. It uses a slim CPU base, runs as a non-root user, downloads the public champion model when it is absent from the build context, and exposes port 8000. Compose defines the web and database services, health checks, readiness dependency, and named volumes. The local Compose run reached healthy status for both services, and the web health endpoint returned the expected model and version response. The production image includes the PostgreSQL driver required by the Compose database configuration.

The first local build exposed two compatibility issues: the installed Gunicorn version did not support the --factory flag, and the database layer used SQLite parameter placeholders with PostgreSQL. The command was changed to Gunicorn's callable-factory syntax, the PostgreSQL driver was added to the serving requirements, and database schema and query placeholders were made backend-aware. The

corrected stack then started successfully.

7. API deployment

The application was deployed from the GitHub main branch on Blitz Cloud using the repository Dockerfile. The deployment uses the public address https://23l-0588-a2-ops.ghauri.prod.page. The build initially failed because the training manifest combined TensorFlow 2.21.0 with MLflow 2.19.0, which require incompatible protobuf ranges. The production and training dependencies were separated, the updated image built successfully, and the public /health endpoint returned the deployed model name, version, and ok status. The free service may sleep when idle; no paid cloud credits were used.

8. Flask and hosted API integration

The deployed Flask service accepts image uploads at /predict, renders the prediction result, and records successful requests for /history. A real dataset image was sent to the public endpoint and returned an AI-generated decision with 71.3% confidence. The health endpoint was checked independently from outside the local workspace and returned the deployed model name, version, and an ok status. The service uses SQLite for the free deployment because no managed database was provisioned; production deployments should use PostgreSQL and a secret-managed configuration.

Submission checklist

• [x] PDF renamed to 23L-0588_VisionGuard.pdf

• [x] GitHub repository link

• [x] DagsHub project link

• [x] Live API URL tested from another network

• [x] Every retained evidence item is a real, captioned screenshot

• [x] No credentials appear in source, screenshots, or report

Appendix A — Saved evidence screenshots

• 01_colab_runtime_t4.png — Colab runtime configuration with a T4 GPU selected.

• 02_colab_secrets_panel.png — Colab Secrets setup panel for DagsHub credentials.

• 10_github_ci_failed.png — Intentional red GitHub Actions run retained for CI lab evidence.

• 15_colab_environment_gpu.png — Colab environment output showing TensorFlow and GPU detection.

• 16_dataset_download_counts.png — Official dataset download path and class file counts.

• 17_dataset_audit_splits.png — Audit count and reproducible train/validation/test split creation.

• 18_dagshub_mlflow_configuration.png — DagsHub MLflow tracking configuration with the token hidden.

• 19_baseline_training_result.png — Baseline CNN training output and evaluation metrics.

• 20_dagshub_final_runs.png — DagsHub MLflow experiment list showing completed VisionGuard runs.

• 21_finetuned_training_result.png — MobileNetV2 fine-tuning output and evaluation metrics.

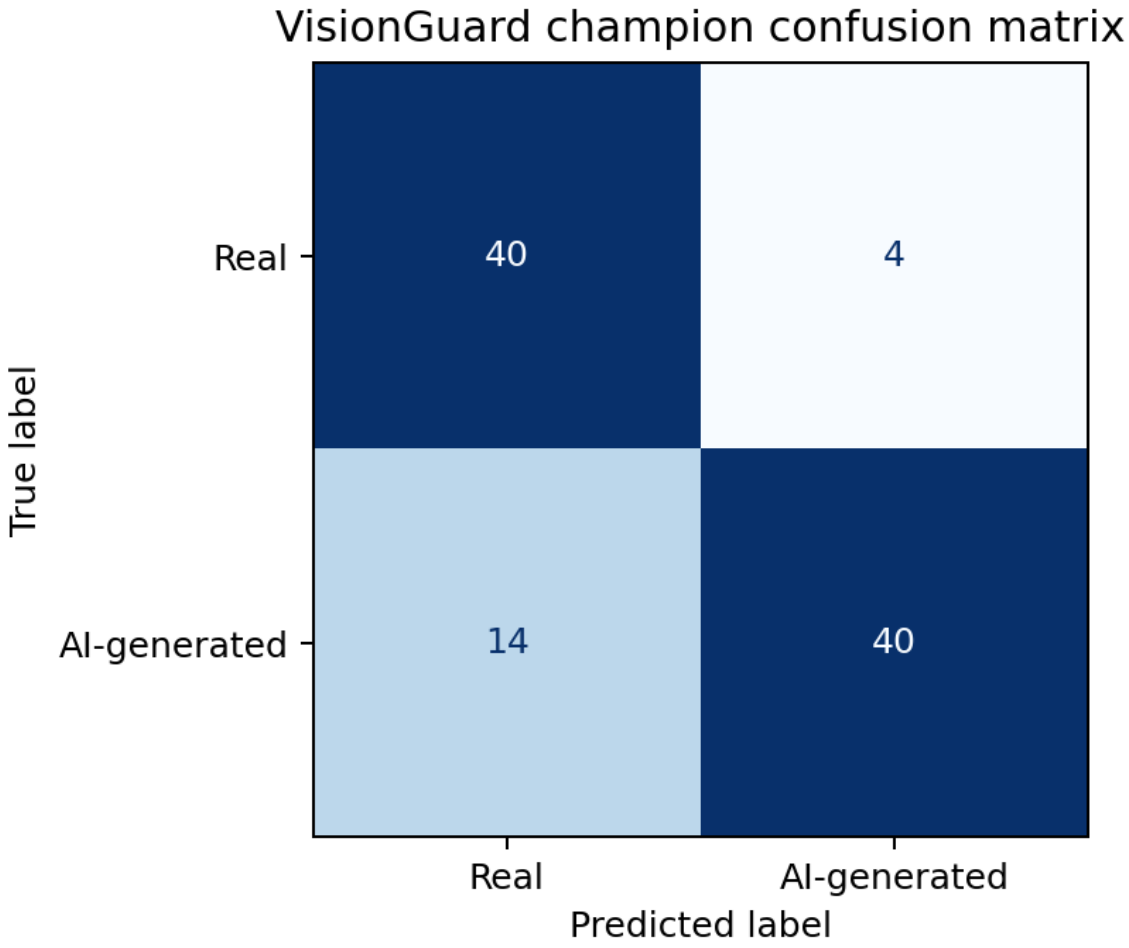
• 22_huggingface_space_files.png — Hugging Face Space file listing captured while diagnosing deployment layout.

• 23_google_cloud_trial_pending.png — Google Cloud free trial pending because prepayment verification was required.

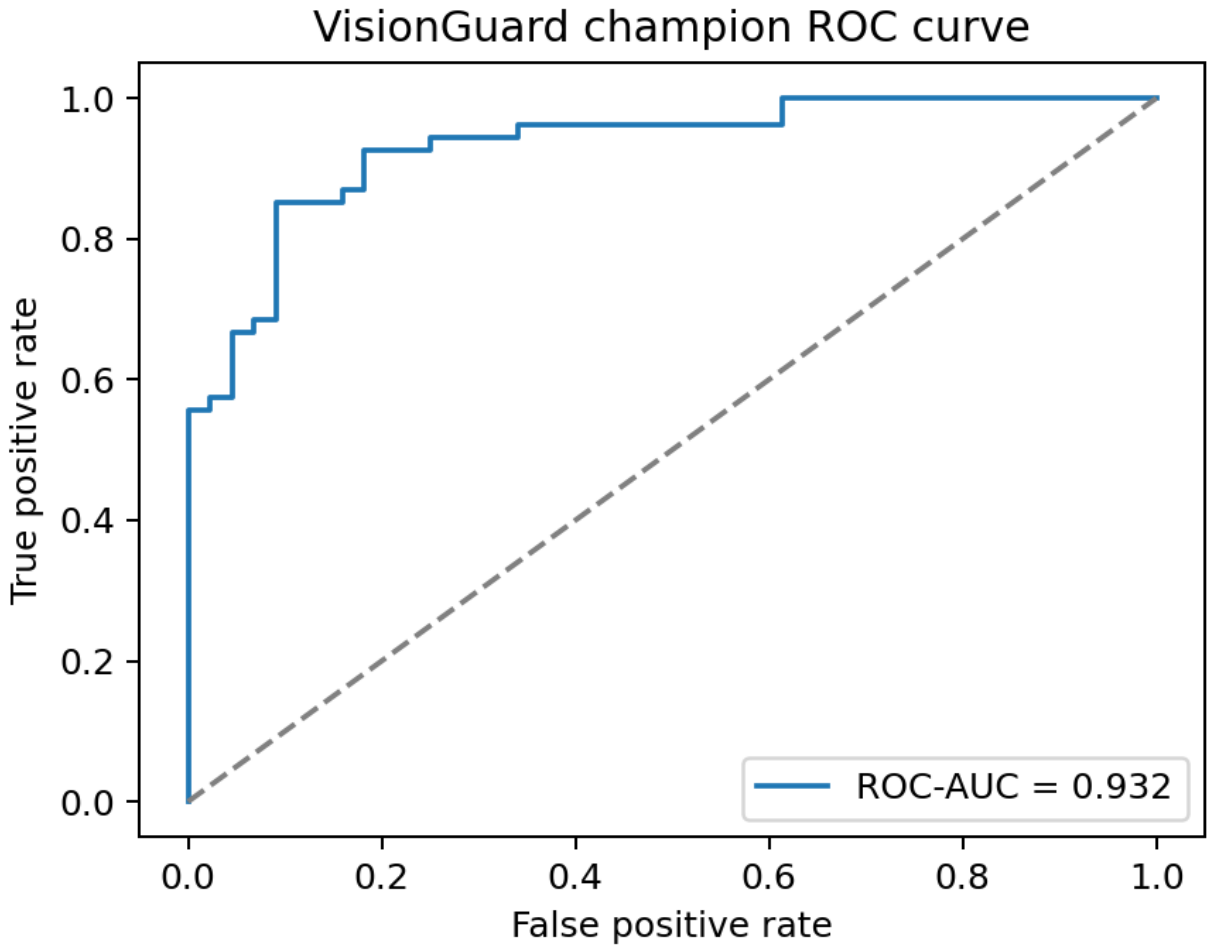
• 24_blitz_dashboard_online.png — Blitz Cloud dashboard showing the public deployment online.

• 25_blitz_public_ui.png — Public Blitz URL displaying the VisionGuard upload interface.

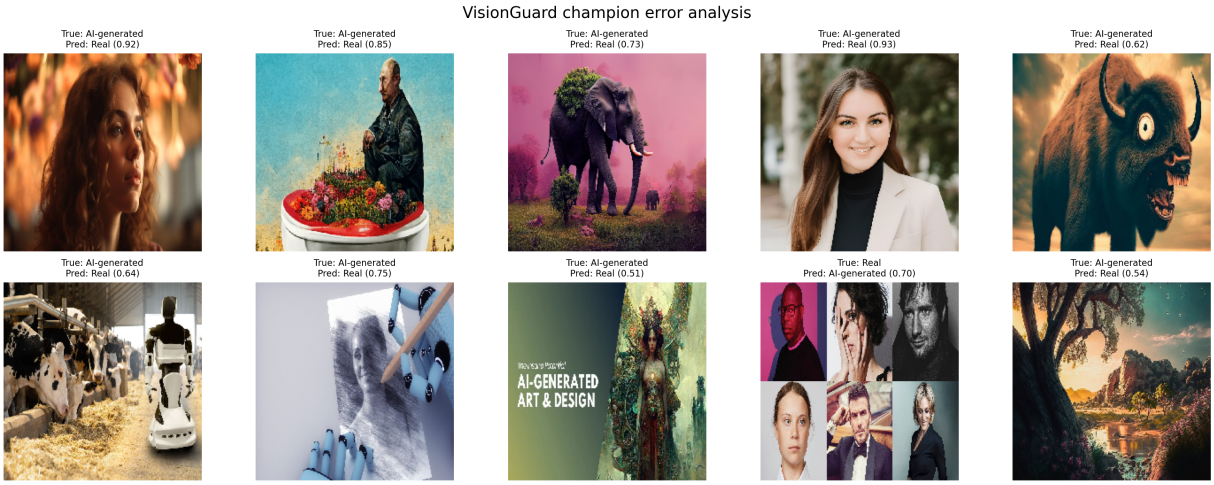
Evaluation figures



confusion_matrix.png *— Champion confusion matrix from the Colab evidence run.*

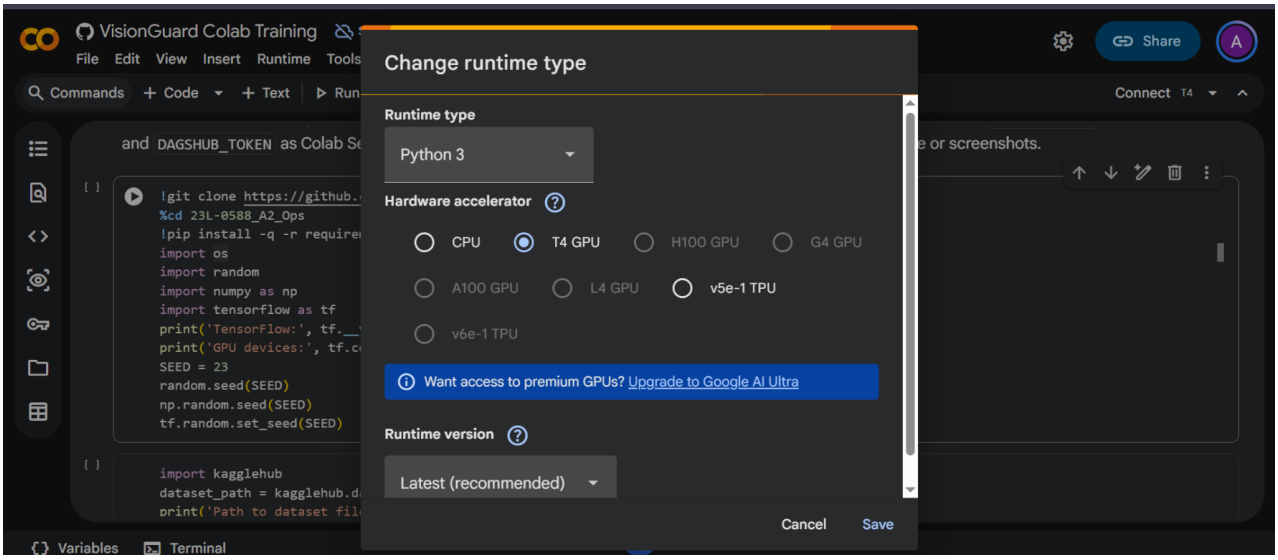


roc_curve.png *— Champion ROC curve and area-under-curve result.*

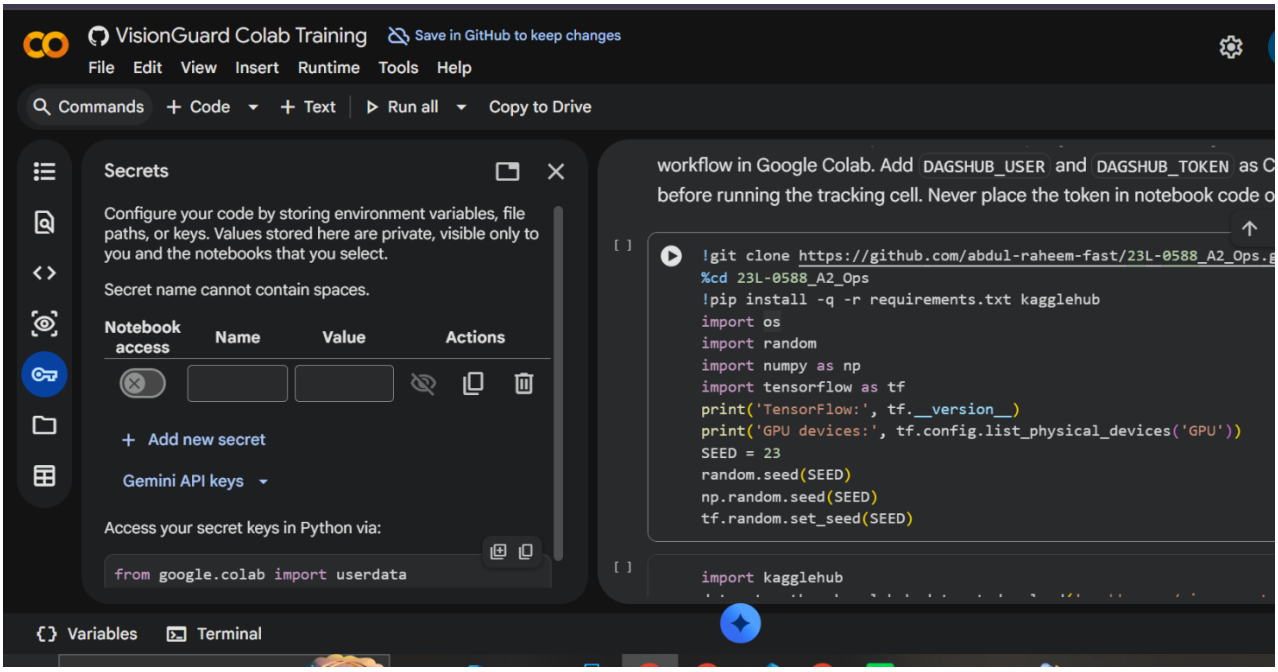


error_analysis_grid.png *— Ten selected test-set errors used for qualitative analysis.*

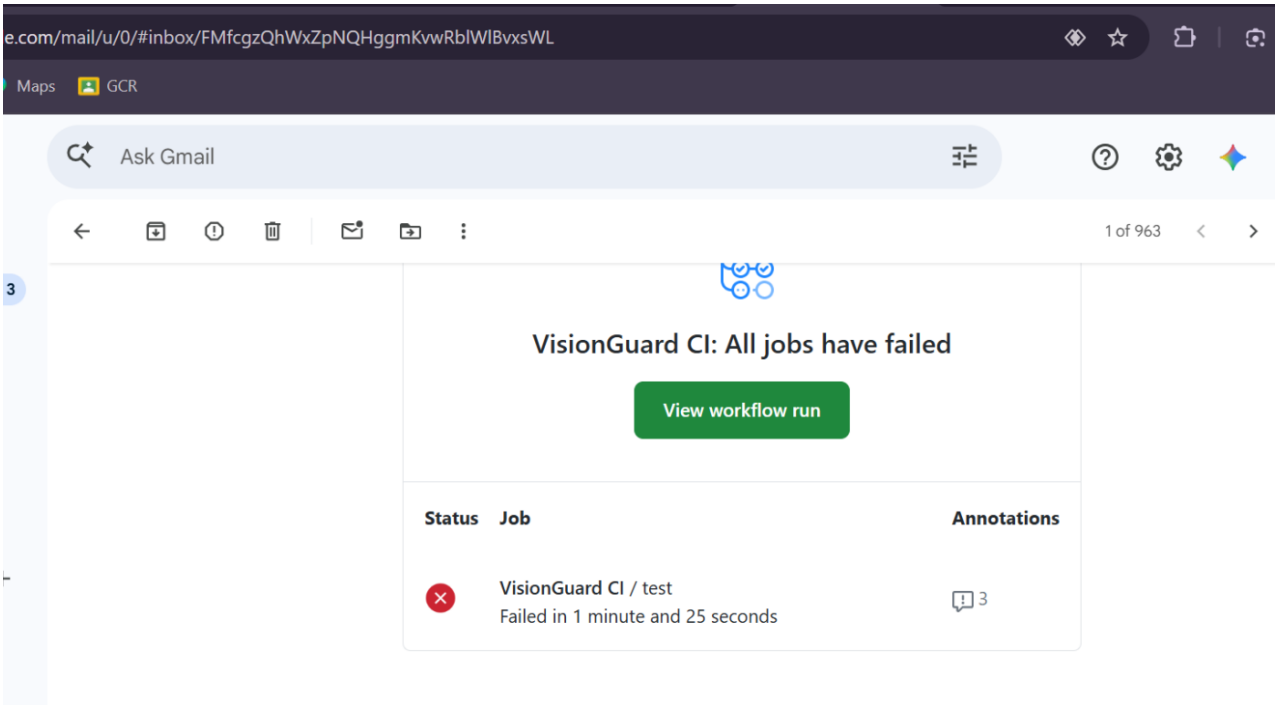
Appendix — Saved evidence screenshots



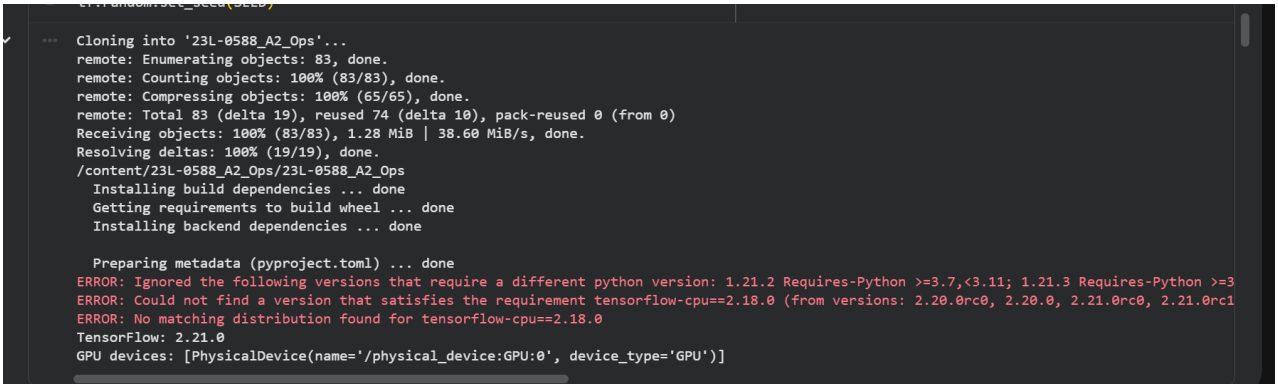
01_colab_runtime_t4.png *— Colab runtime configuration with a T4 GPU selected.*



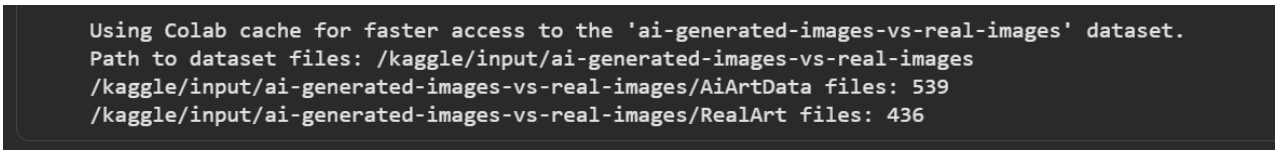
02_colab_secrets_panel.png *— Colab Secrets setup panel for DagsHub credentials.*



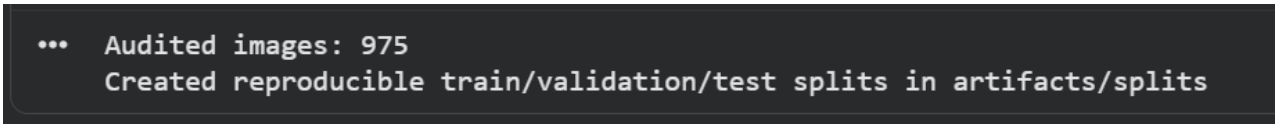
10_github_ci_failed.png *— Intentional red GitHub Actions run retained for CI lab evidence.*



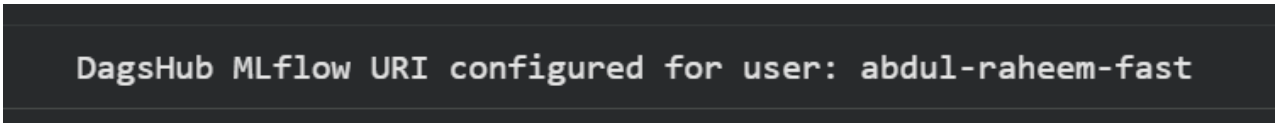
15_colab_environment_gpu.png *— Colab environment output showing TensorFlow and GPU detection.*



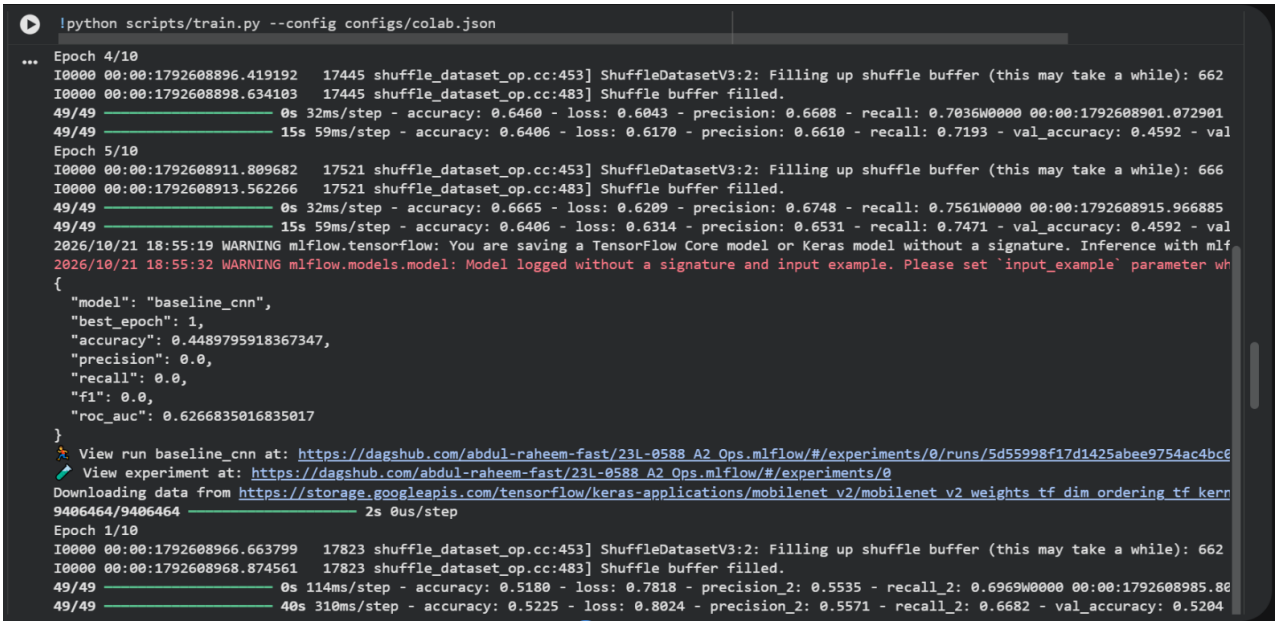
16_dataset_download_counts.png *— Official dataset download path and class file counts.*



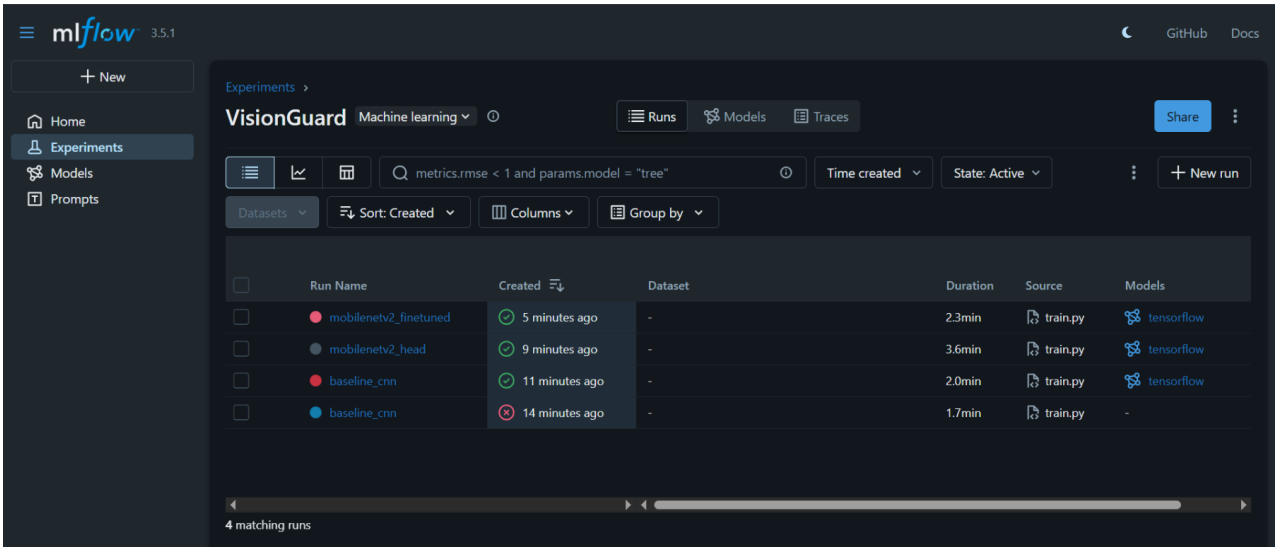
17_dataset_audit_splits.png *— Audit count and reproducible train/validation/test split creation.*



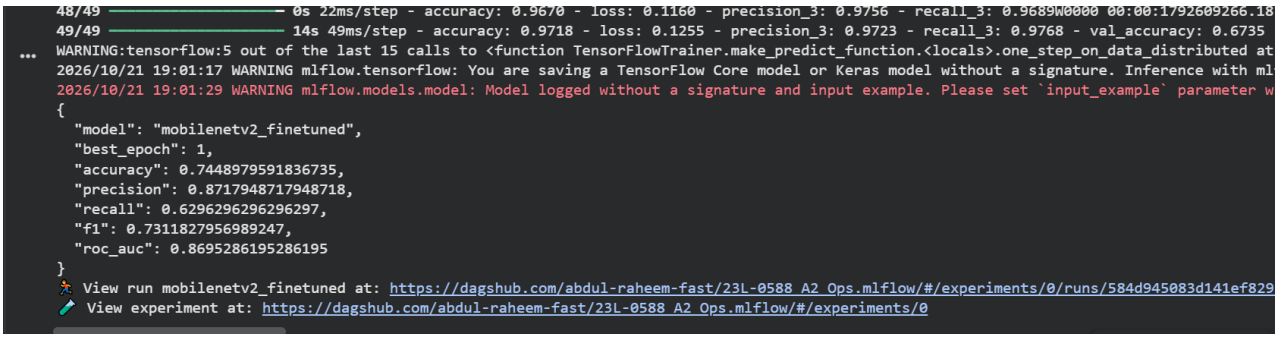
18_dagshub_mlflow_configuration.png *— DagsHub MLflow tracking configuration with the token hidden.*



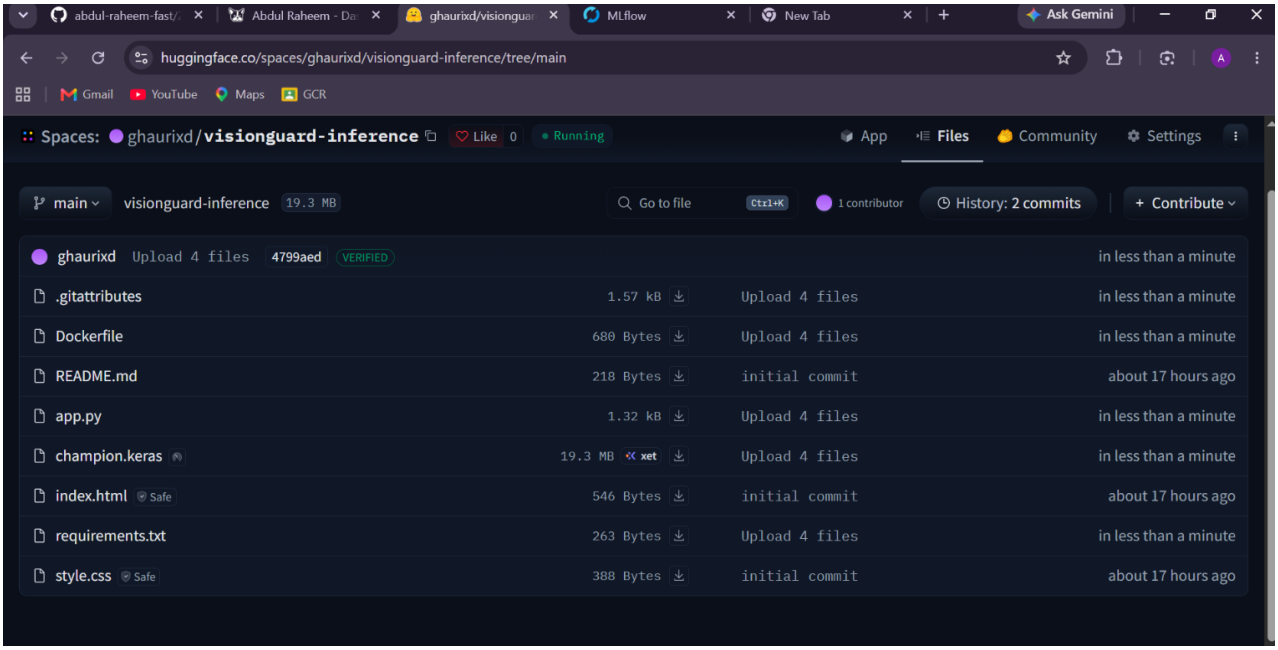
19_baseline_training_result.png *— Baseline CNN training output and evaluation metrics.*



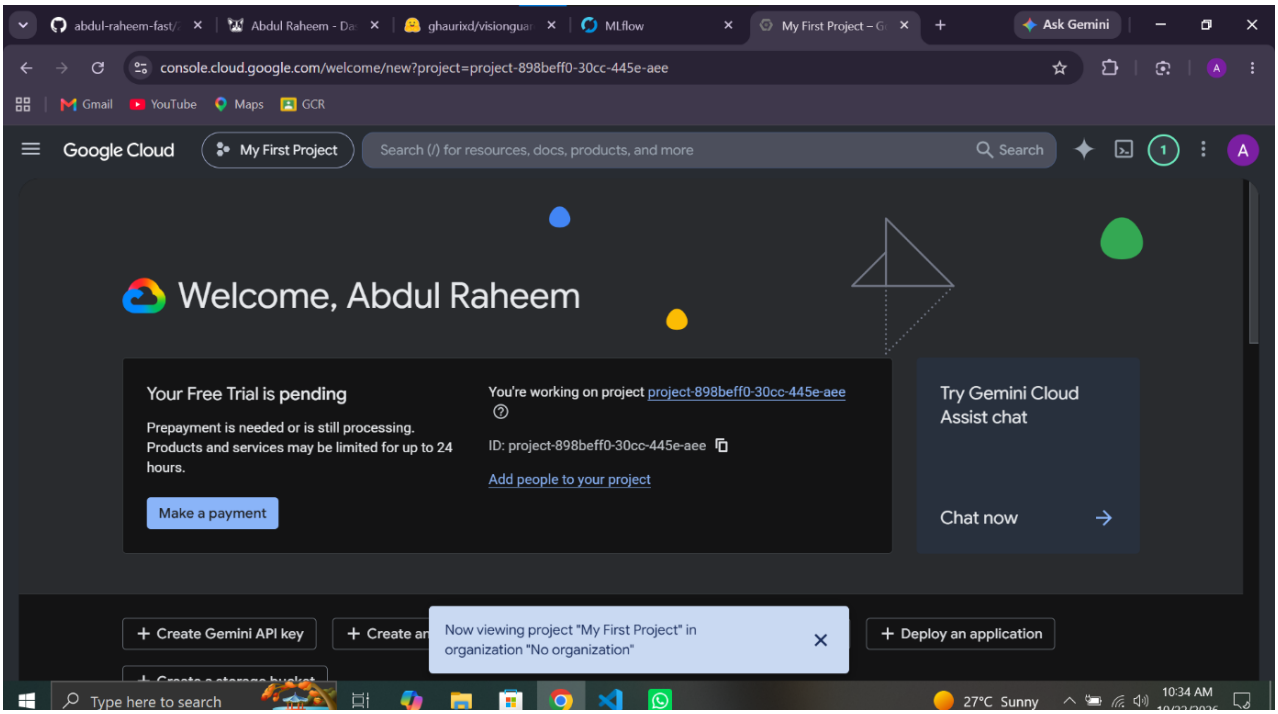
20_dagshub_final_runs.png *— DagsHub MLflow experiment list showing completed VisionGuard runs.*



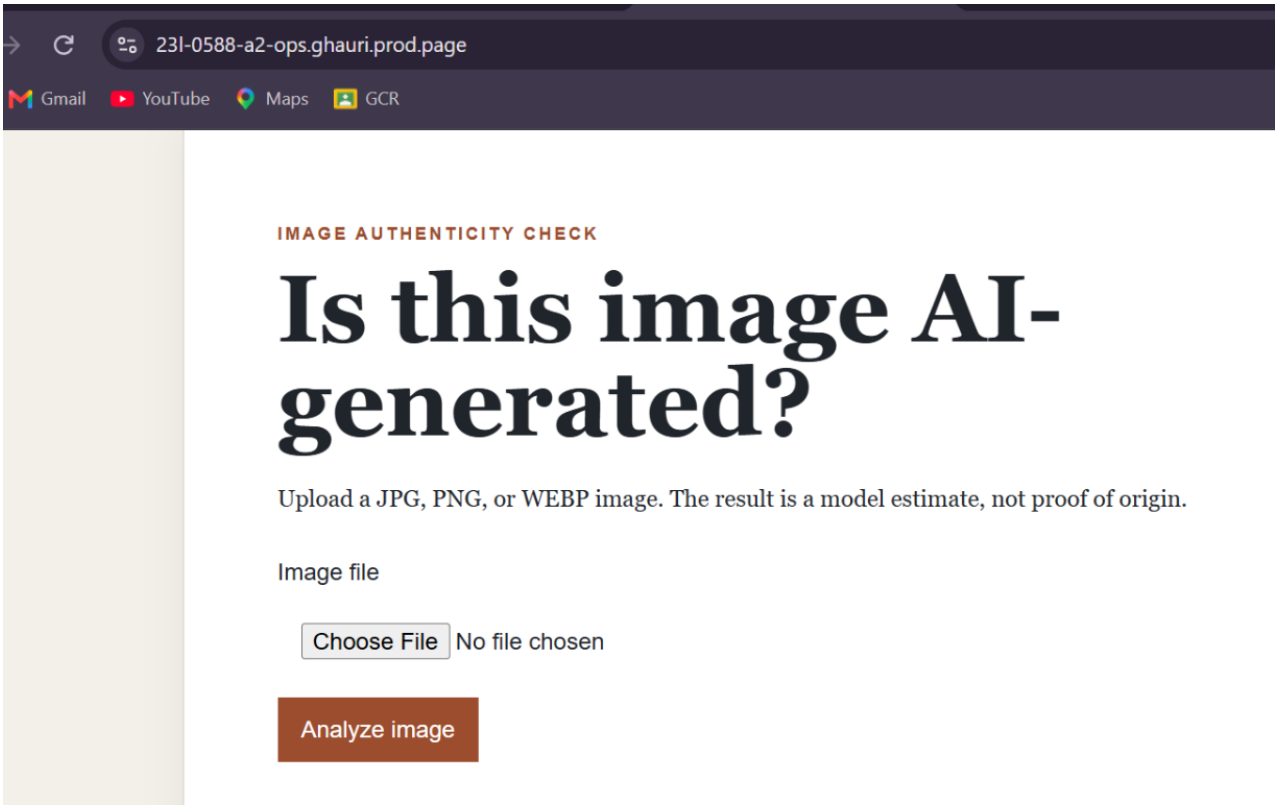
21_finetuned_training_result.png *— MobileNetV2 fine-tuning output and evaluation metrics.*



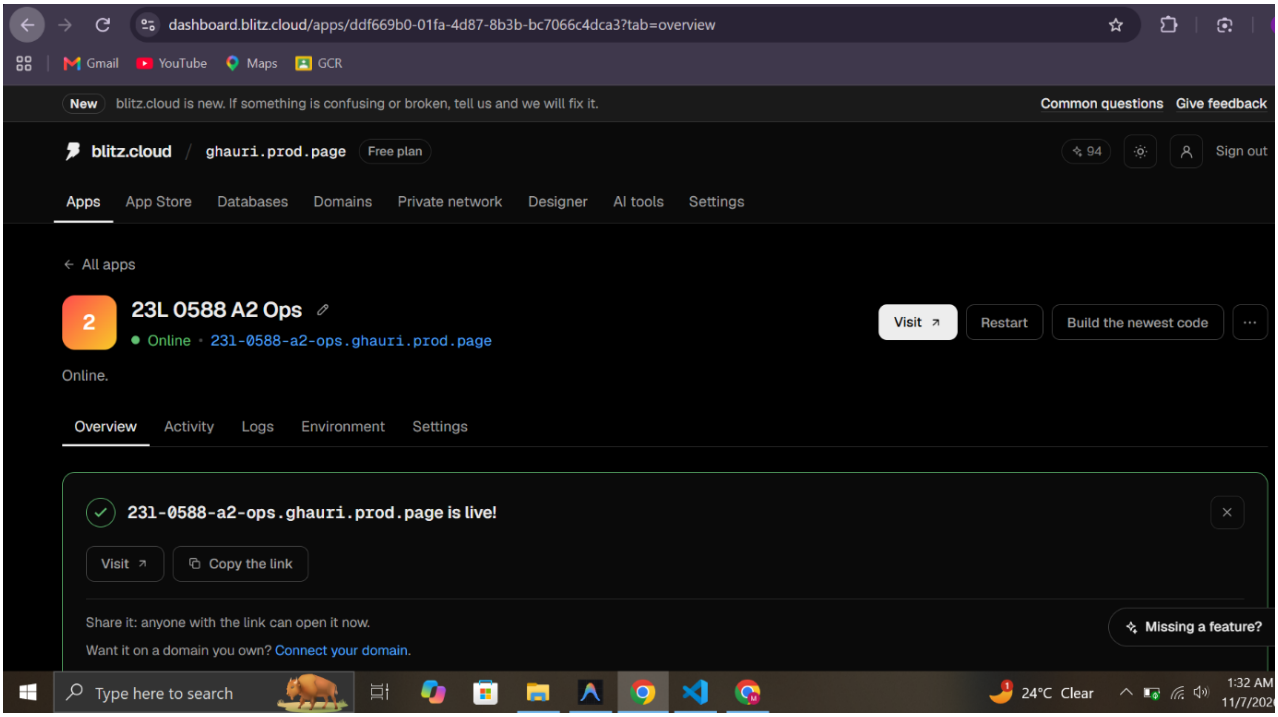
22_huggingface_space_files.png *— Hugging Face Space file listing captured while diagnosing the deployment layout.*



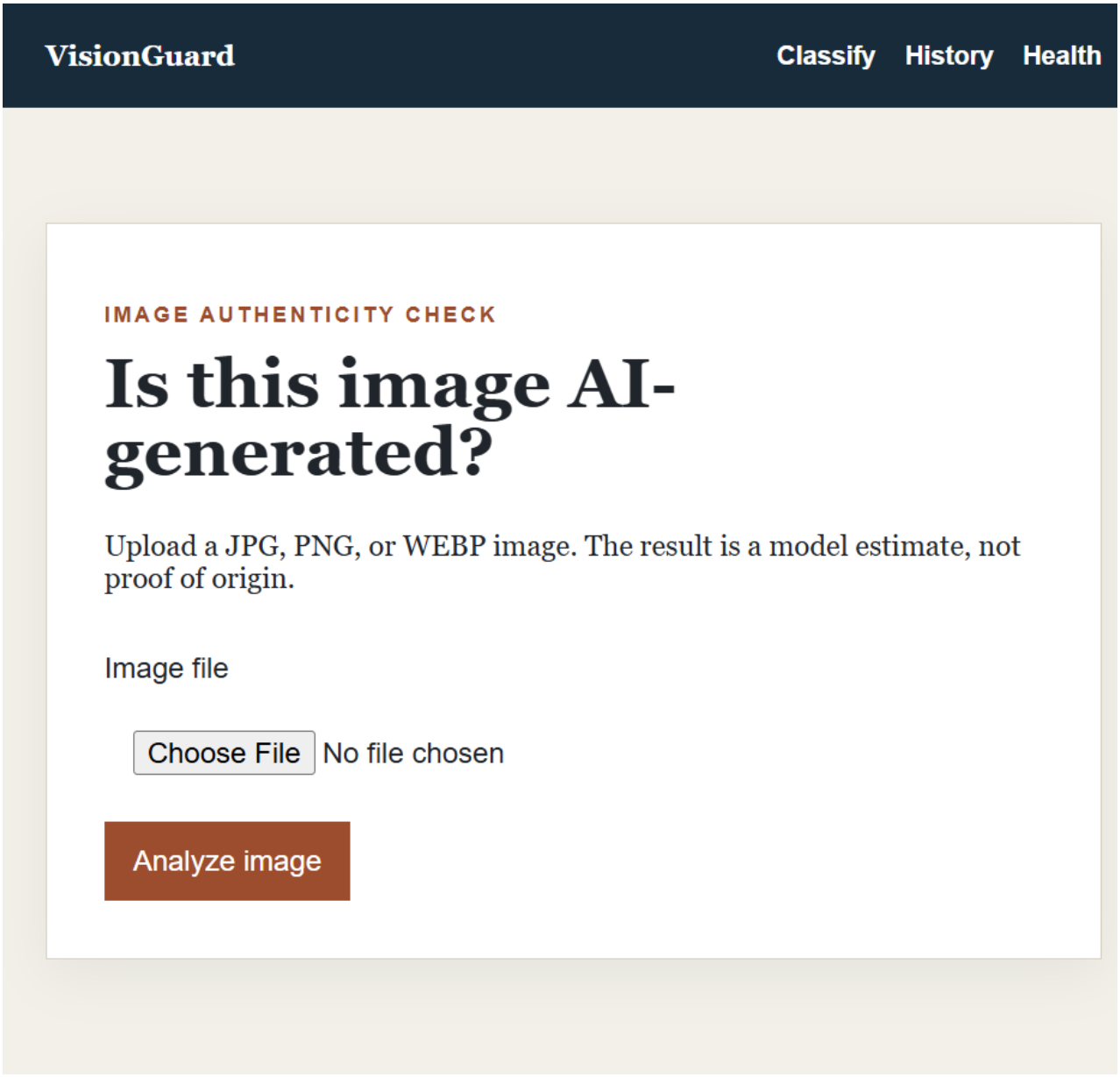
23_google_cloud_trial_pending.png *— Google Cloud free trial pending because prepayment verification was required.*



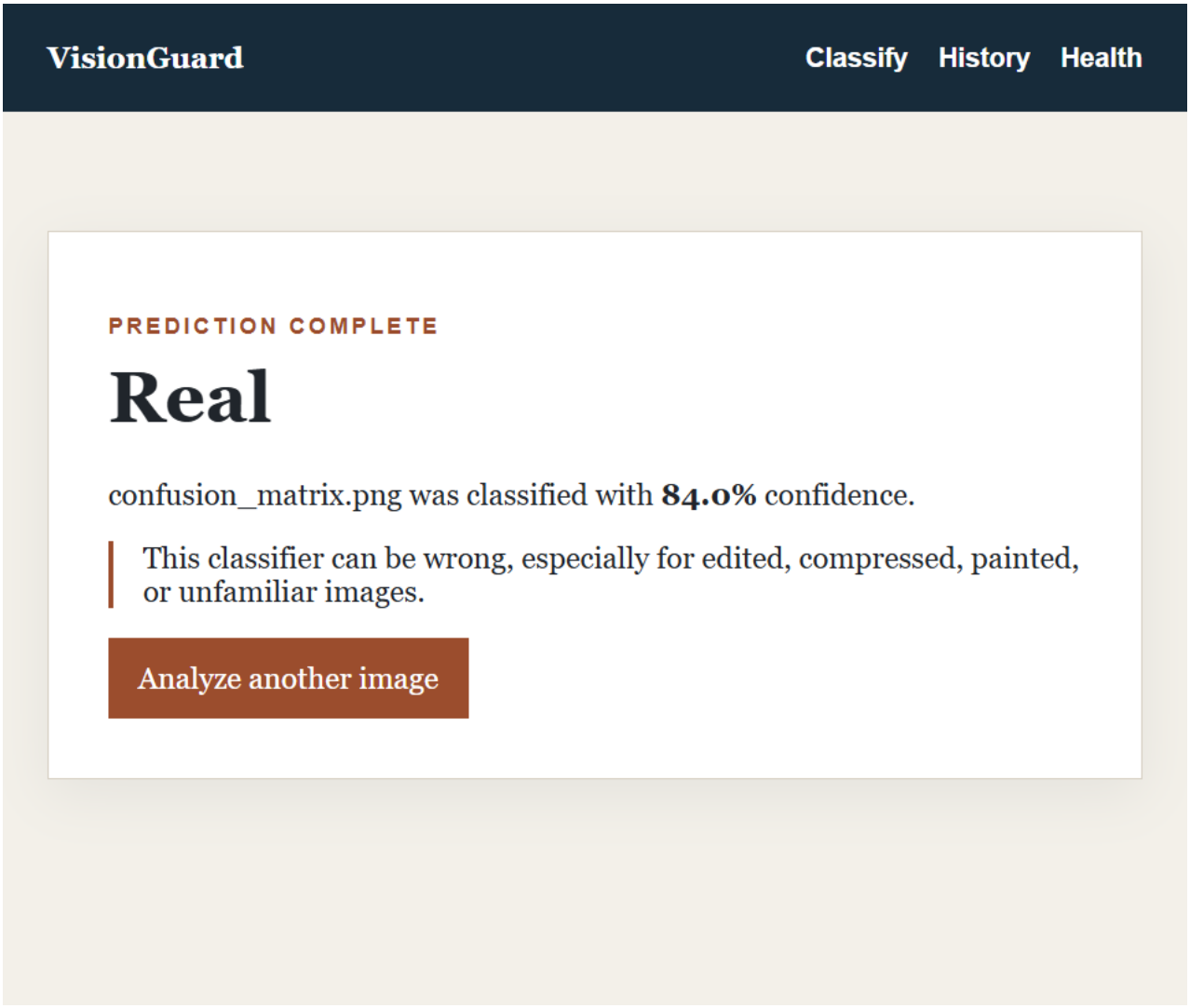
24_blitz_dashboard_online.png *— Blitz Cloud dashboard showing the deployed application online with its public address.*



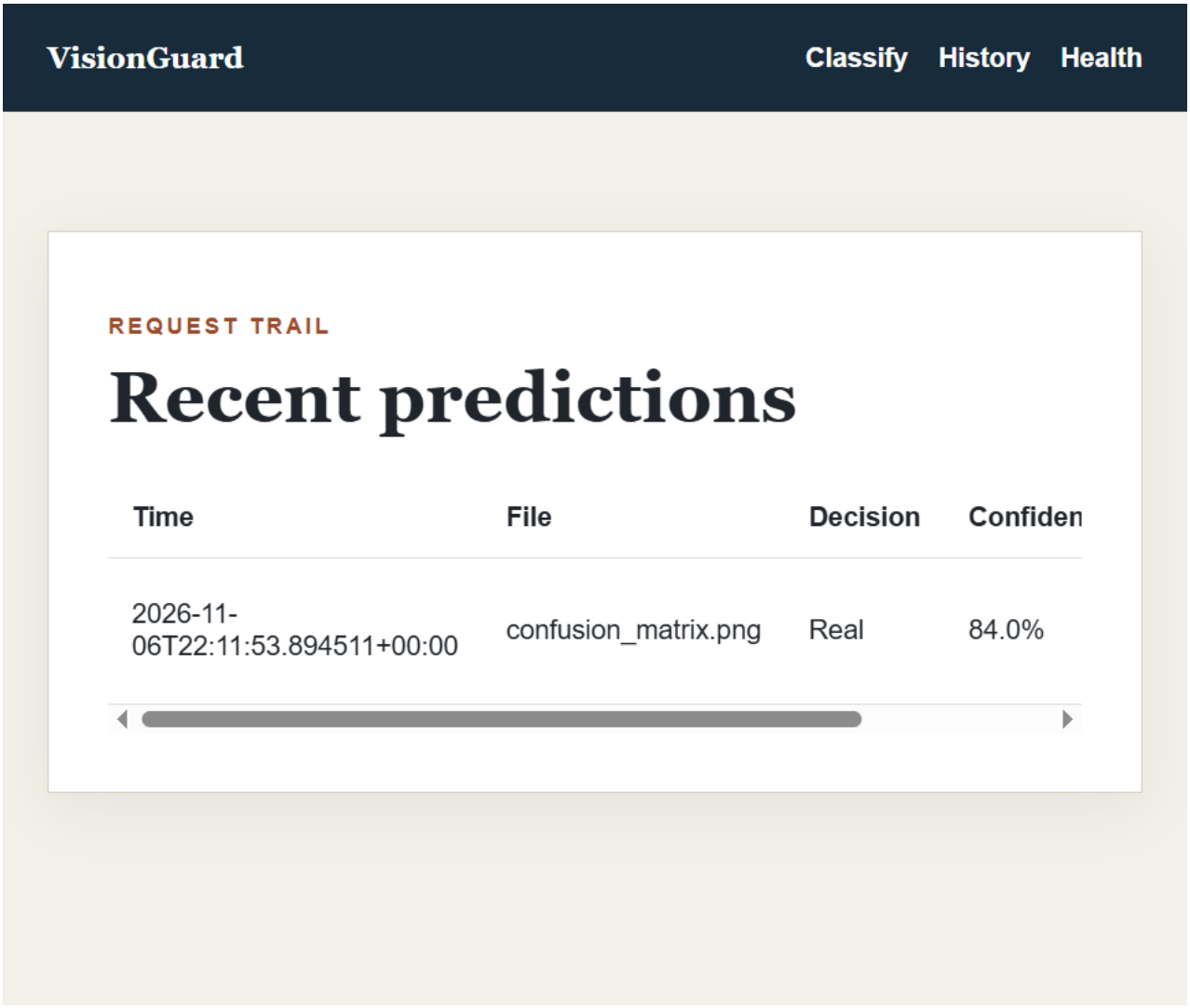
25_blitz_public_ui.png *— Public Blitz URL displaying the VisionGuard upload interface.*



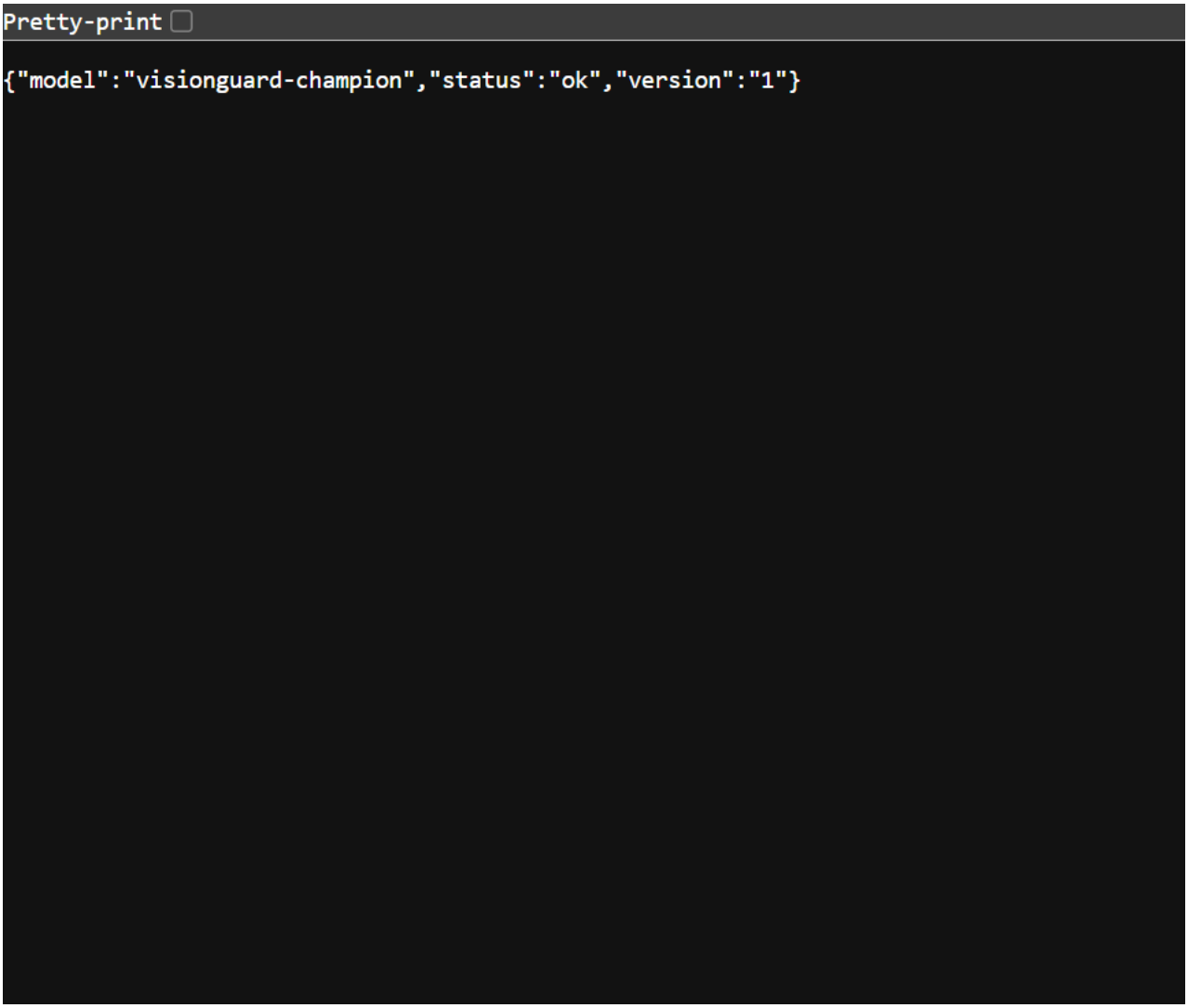
26_compose_local_ui.png *— Tasks 4.2, 4.3, and 6.3 — local Docker Compose interface with the upload, history, and health routes.*



27_compose_prediction_result.png *— Tasks 4.1, 4.2, and 8.1 — real local prediction result showing the label, confidence, and model caveat.*

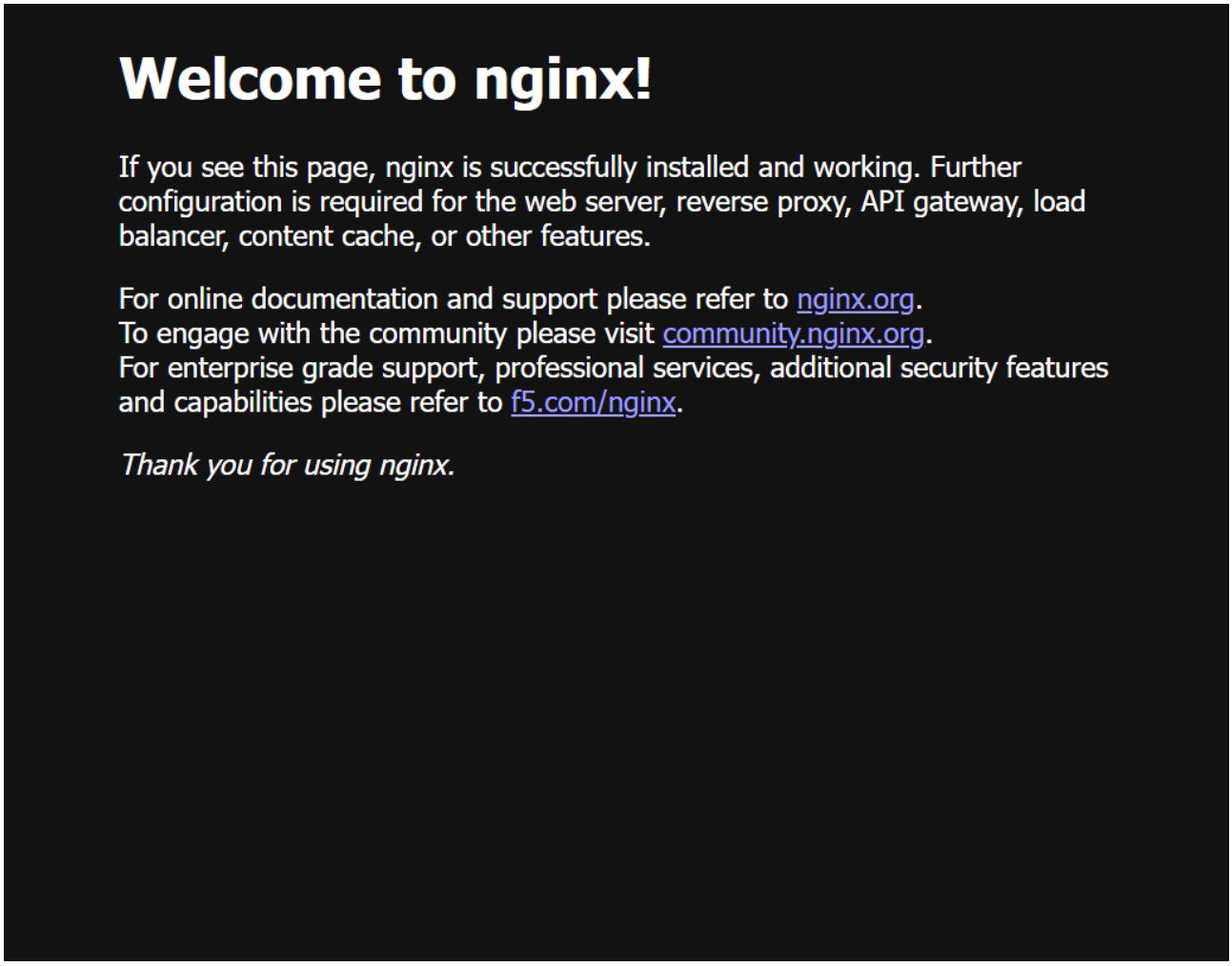


28_compose_history.png *— Tasks 5 and 8.1 — database-backed history showing timestamp, filename, decision, confidence, model, and version.*

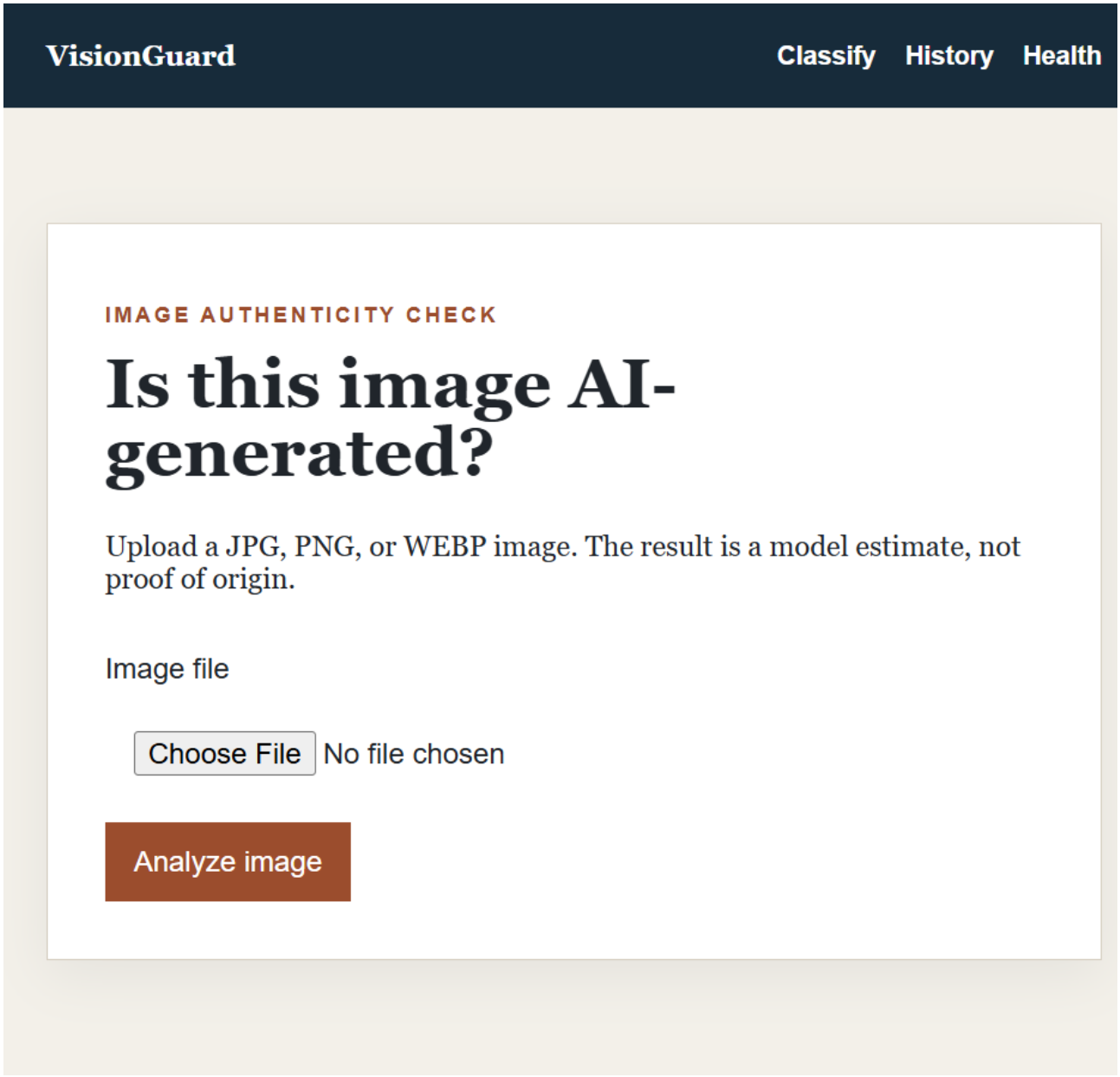


29_compose_health.png *— Tasks 4.4 and 6.3 — health response from the Compose web service showing model, version, and status.*

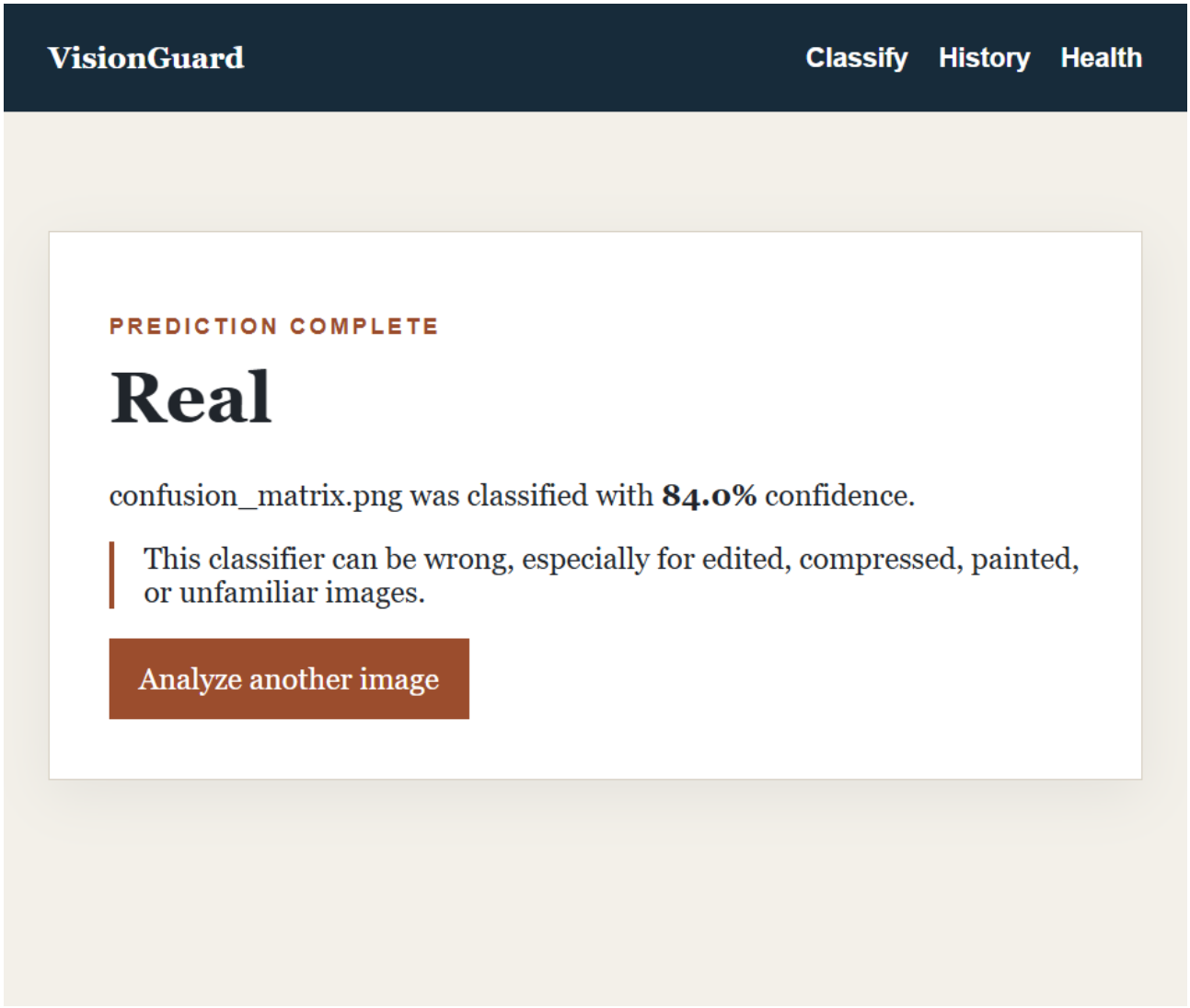
Additional Docker Evidence



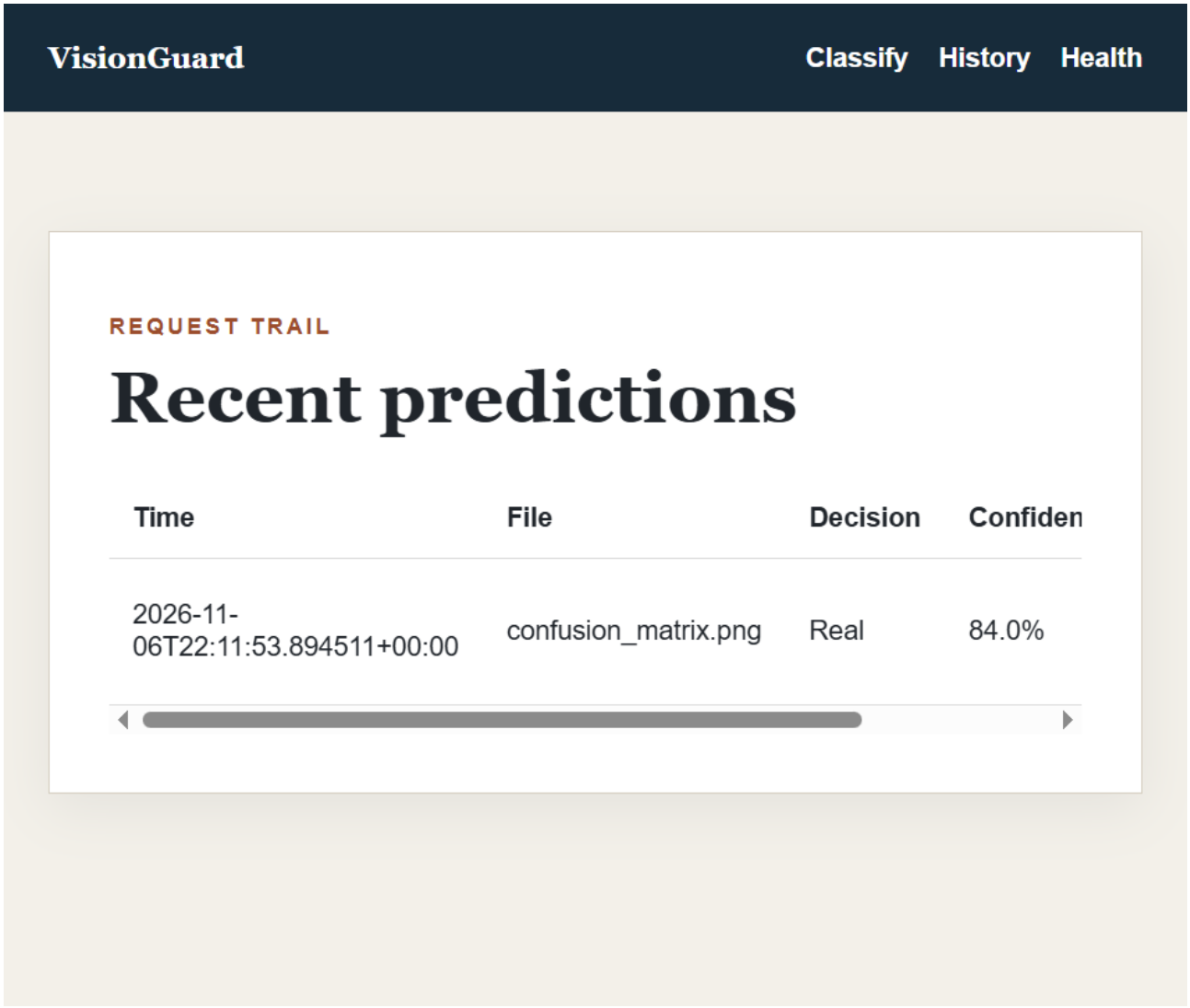
6.1 — Nginx warm-up container running on localhost:18080.



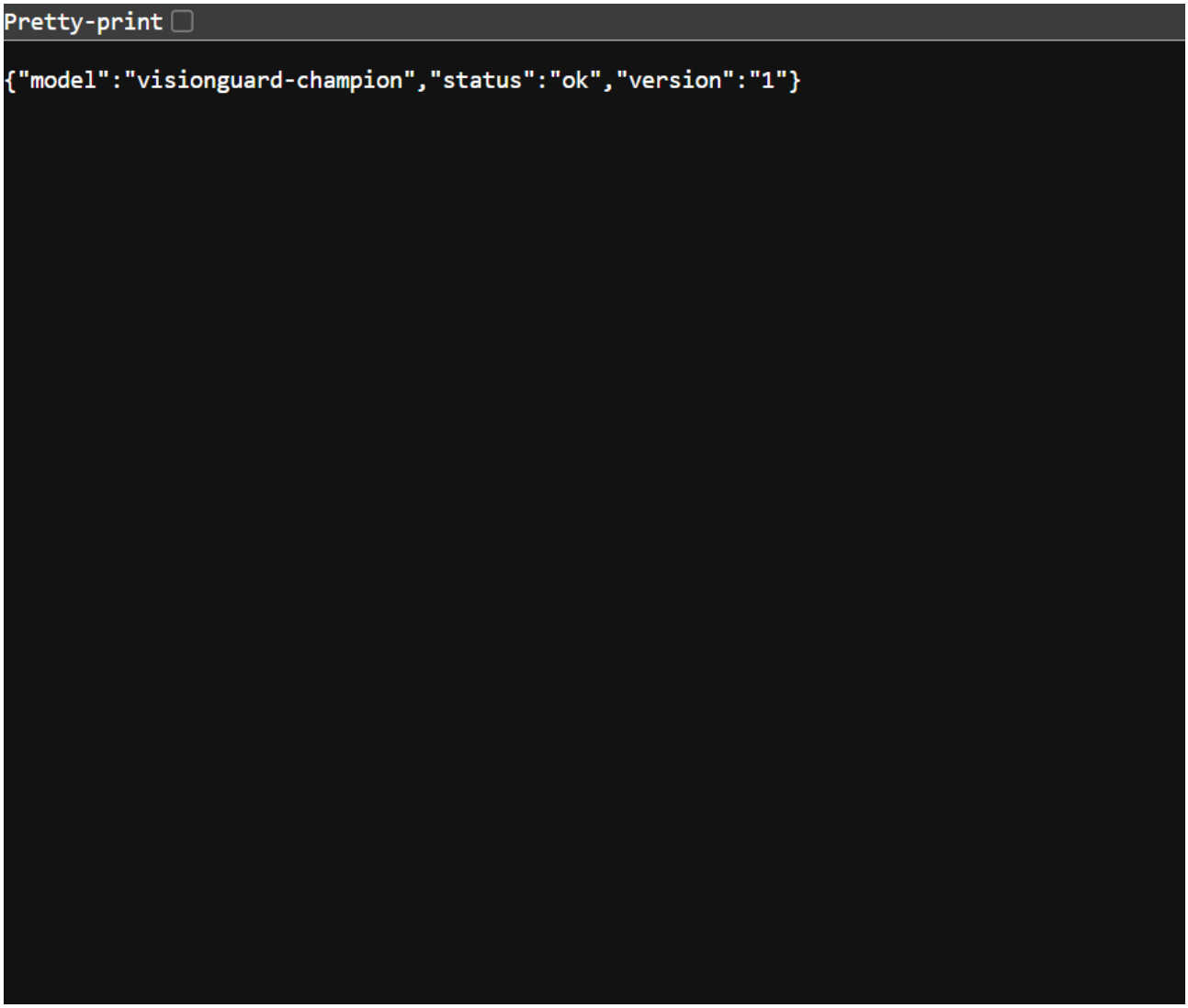
4.2, 4.3, 6.3 — Local Compose upload interface.



4.1, 4.2, 8.1 — Local Compose prediction result.



5, 8.1 — Database-backed prediction history.



4.4, 6.3 — Local Compose health response.

docker-warmup/01_run_nginx.txt

```
■■0■2■f■4■f■e■6■a■d■c■6■c■3■a■1■b■5■e■9■7■6■f■5■3■1■e■8■9■2■4■9■6■d■a■8■4■d■5■6■0■0■
f■b■7■f■6■8■2■f■f■5■e■a■1■8■6■8■a■d■1■a■5■3■a■
■
■
```

docker-warmup/02_docker_ps.txt

```
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
0f2ff4fe6adc6c


cucurrencyc-ept-lt-pipeline_2_9b6e5-web-1

1f7c1c3c1107f9


cucurrencyc-ept-lt-pipeline_2_9b6e5-worker-1

1d4626d2d6db


cucurrencyc-ept-lt-pipeline_2_9b6e5-tracing-1

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cucurrencyc-ept-lt-pipeline_2_9b6e5-mock-1

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cucurrencyc-ept-lt-pipeline_2_9b6e5-postgres-1

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cucurrencyc-ept-lt-pipeline_2_9b6e5-valkey-1

f3bb4bbecc2cce


cucurrencyc-ept-lt-pipeline_2_9b6e5-postgres-1

5432/tcp

vkeyc-ept-lt-pipeline_2_9b6e5-mock-1
```

docker-warmup/03_docker_logs.txt

```
■■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■d■/■ ■i■s■ ■n■o■t■ ■e■m■p■t■y■,■ ■w■i■l■l■
■a■t■t■e■m■p■t■ ■t■o■ ■p■e■r■f■o■r■m■ ■c■o■n■f■i■g■u■r■a■t■i■o■n■
■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■ ■L■o■o■k■i■n■g■ ■f■o■r■ ■s■h■e■l■l■
■s■c■r■i■p■t■s■ ■i■n■ ■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■d■/■
■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■ ■L■a■u■n■c■h■i■n■g■ ■/■d■o■c■k■e■r■-■e
■n■t■r■y■p■o■i■n■t■.■d■/■1■0■-■l■i■s■t■e■n■-■o■n■-■i■p■v■6■-■b■y■-■d■e■f■a■u■l■t■.■s
■h■
■
■1■0■-■l■i■s■t■e■n■-■o■n■-■i■p■v■6■-■b■y■-■d■e■f■a■u■l■t■.■s■h■:■ ■i■n■f■o■:■
■G■e■t■t■i■n■g■ ■t■h■e■ ■c■h■e■c■k■s■u■m■ ■o■f■
■/■e■t■c■/■n■g■i■n■x■/■c■o■n■f■.■d■/■d■e■f■a■u■l■t■.■c■o■n■f■
■
■1■0■-■l■i■s■t■e■n■-■o■n■-■i■p■v■6■-■b■y■-■d■e■f■a■u■l■t■.■s■h■:■ ■i■n■f■o■:■
■E■n■a■b■l■e■d■ ■l■i■s■t■e■n■ ■o■n■ ■I■P■v■6■ ■i■n■
■/■e■t■c■/■n■g■i■n■x■/■c■o■n■f■.■d■/■d■e■f■a■u■l■t■.■c■o■n■f■
■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■ ■S■o■u■r■c■i■n■g■ ■/■d■o■c■k■e■r■-■e■n
■t■r■y■p■o■i■n■t■.■d■/■1■5■-■l■o■c■a■l■-■r■e■s■o■l■v■e■r■s■.■e■n■v■s■h■
■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■ ■L■a■u■n■c■h■i■n■g■ ■/■d■o■c■k■e■r■-■e
■n■t■r■y■p■o■i■n■t■.■d■/■2■0■-■e■n■v■s■u■b■s■t■-■o■n■-■t■e■m■p■l■a■t■e■s■.■s■h■
■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■ ■L■a■u■n■c■h■i■n■g■ ■/■d■o■c■k■e■r■-■e
■n■t■r■y■p■o■i■n■t■.■d■/■3■0■-■t■u■n■e■-■w■o■r■k■e■r■-■p■r■o■c■e■s■s■e■s■.■s■h■
■
■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■s■h■:■ ■C■o■n■f■i■g■u■r■a■t■i■o■n■
■c■o■m■p■l■e■t■e■;■ ■r■e■a■d■y■ ■f■o■r■ ■s■t■a■r■t■ ■u■p■
■
■d■o■c■k■e■r■ ■:■ ■2■0■2■6■/■1■1■/■0■7■ ■0■7■:■1■2■:■5■8■ ■[■n■o■t■i■c■e■]■ ■1■#■1■:■
■u■s■i■n■g■ ■t■h■e■ ■"■e■p■o■l■l■"■ ■e■v■e■n■t■ ■m■e■t■h■o■d■
■
■A■t■ ■l■i■n■e■:■2■ ■c■h■a■r■:■3■0■9■
■
■+■ ■.■.■.■ ■s■ ■|■ ■T■e■e■-■O■b■j■e■c■t■
■"■$■o■u■t■\■0■2■_■d■o■c■k■e■r■_■p■s■.■t■x■t■"■;■ ■d■o■c■k■e■r■ ■l■o■g■s■ ■w■e■b■1■
■2■>■&■1■ ■|■ ■T■e■e■-■O■ ■.■.■.■
■
■+■ ■ ■ 
```

docker-warmup/04_exec_ls.txt

```
■■5■0■x■.■h■t■m■l■
■
■i■n■d■e■x■.■h■t■m■l■
■
■
```

docker-warmup/05_docker_ps_a.txt

```
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

0f2f4fe6ad6c6c   nginx:1.18.0-alpine   "/docker-entrypoint.…"   About a minute ago   Up 3 seconds   80/tcp   sleepy_poitras
4   sse cond s   agno   U p   3   sse cond s
0.0.0.0:18080->80/tcp,   [::]:18080->80/tcp
   web1

ab9665507d9760   cuurrenncy-eetll-ppippellinee_29b66e-web1-1
5/mailrflowv:llatessts   "/dockerntrypoint"   About a minute ago   Up 2   hourrs   agno   80/tcp
2   hourrs
1273.0.0.1:8080->80/tcp   cuurrenncy-eetll-ppippellinee_29b66e-web1-1

1f73c1c3107095   cuurrenncy-eetll-ppippellinee_29b66e-web1-1
5/mailrflowv:llatessts   "/dockerntrypoint"   About a minute ago   Up 2   hourrs   agno
2   hourrs
cuurrenncy-eetll-ppippellinee_29b66e-web1-1

e61d4626226d6dbb   cuurrenncy-eetll-ppippellinee_29b66e-web1-1
5/mailrflowv:llatessts   "/dockerntrypoint"   About a minute ago   Up 2   hourrs   agno
2   hourrs
cuurrenncy-eetll-ppippellinee_29b66e-trriggger-1

1e12e1e66d6df0fcc   mogno-exprressss:1.0   "/sbinn/ttinni   About a minute ago   Up 2   hourrs   agno
2   hourrs
0.0.0.0:8081->1/tcp,
[::]:8081->1/tcp   cuurrenncy-eetll-ppippellinee_29b66e-mogno-1

6cd09b9b94883fdd   ppossttgrreess:1.15   "/docker-entrypoint.…"   About a minute ago   Up 2   hourrs   agno   80/tcp
2   hourrs
1273.0.0.1:5432->5432/tcp   cuurrenncy-eetll-ppippellinee_29b66e-ppossttgrreess-1

c7bb3d7a40a03000   mogno:7.0   "docker-entrypoint.s…"   About a minute ago   Up 2   hourrs   agno   80/tcp
2   hourrs
0.0.0.0:27017->27017/tcp,
[::]:27017->27017/tcp   cuurrenncy-eetll-ppippellinee_29b66e-mogno-1

16b3e72b94a23bb4   vilisiongunarrd-web1   "/docker-entrypoint.…"   About a minute ago   Up 2   hourrs   agno   80/tcp
9   hourrs   (healthy)
0.0.0.0:8000->80/tcp,
[::]:8000->80/tcp   vilisiongunarrd-web1-1

f3bb4bbec2cce3f0   ppossttgrreess:16-alllpiinee   "docker-entrypoint.s…"   About a minute ago   Up 2   hourrs   agno   80/tcp
9   hourrs   (healthy)   5432/tcp   cuurrenncy-eetll-ppippellinee_29b66e-ppossttgrreess-1
54321/tcp   vilisiongunarrd-ddattabaasee-1

```

docker-warmup/06_docker_stop.txt

```
■■w■e■b■1■
■
■
```

docker-warmup/07_docker_rm.txt

```
■■w■e■b■1■
■
■
```

docker-warmup/08_docker_images.txt

```
REPOSITORY                                                      TAG                 IMAGE ID            CREATED             SIZE
<none>                                                          <none>              d3a7f9b6e5c1        2 days ago          1.35GB
cruderrr/encyber-etl-pipeline_29b6e5c/main:latest               latest              b4e2d3d7e6f02       2 days ago          1.35GB
<none>                                                          <none>              1b23d7e6a1a6f       2 days ago          1.18GB
mongo:7.0                                                       7.0                 8bbb4424e8757       2 days ago          1.18GB
nginx:latest                                                    latest              1665dddde8751       2 days ago          281MB
postgres:16-alpine                                              16-alpine           1aaac5b4c9ce0       2 days ago          395MB
python:3.11-slim                                                3.11-slim           a4a6cbb33cf4c       2 days ago          131MB
visionsinmotion/ubuntu-dev-web:latest                           latest              ee0bb3dd1c556       2 days ago          1.24GB
```

docker-warmup/09_docker_rmi.txt

```
■■U■n■t■a■g■g■e■d■:■ ■n■g■i■n■x■:■l■a■t■e■s■t■
■
■D■e■l■e■t■e■d■:■ ■s■h■a■2■5■6■:■1■6■5■d■d■d■e■8■7■5■1■e■7■5■a■1■c■9■0■7■d■2■4■e■f■1
■0■d■9■f■6■d■e■b■0■8■b■5■7■e■0■f■e■5■6■c■a■f■6■7■5■1■2■4■8■6■a■f■f■7■6■d■8■2■
■
■
```

docker-warmup/11_volume_ps.txt



docker-warmup/12_default_nginx_ps.txt

```
■■C■O■N■T■A■I■N■E■R■ ■I■D■ ■ ■ ■I■M■A■G■E■ ■ ■ ■ ■ ■C■O■M■M■A■N■D■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■ ■ ■ ■ ■ ■C■R■E■A■T■E■D■ ■ ■ ■ ■ ■ ■ ■ ■S■T■A■T■U■S■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■P■O■R■T■S■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■ ■N■A■M■E■S■
■
■8■6■8■2■6■1■2■c■b■d■2■6■ ■ ■ ■n■g■i■n■x■ ■ ■ ■ ■
■"■/■d■o■c■k■e■r■-■e■n■t■r■y■p■o■i■n■t■.■& "■ ■ ■ ■1■ ■s■e■c■o■n■d■ ■a■g■o■ ■ ■ ■U■p■
■L■e■s■s■ ■t■h■a■n■ ■a■ ■s■e■c■o■n■d■ ■ ■
■0■.■0■.■0■.■0■:■1■8■0■8■0■-■>■8■0■/■t■c■p■,■ ■[■:■:■]■:■1■8■0■8■0■-■>■8■0■/■t■c■p■
■ ■ ■w■e■b■1■
■
■
```

docker-ops/01_compose_ps_healthy.txt

```
NAME   IMAGE   SERVICE   COMMAND   STATE   PORTS
visionguard-dataset-watcher-1   visionguard-dataset-watcher   dataset-watcher   "docker-entrypoint.s…"   dataset-watcher   2 hours ago   Up 2 hours (healthy)   0.0.0.0:8000->8000/tcp,
[::]:8000->8000/tcp
visionguard-web-1   visionguard-web   web   "/docker-entrypoint.…"   web   2 hours ago   Up 2 hours (healthy)   0.0.0.0:3000->80/tcp,
[::]:3000->80/tcp
```

docker-ops/02_database_port.txt

```
■■d■o■c■k■e■r■ ■p■o■r■t■ ■v■i■s■i■o■n■g■u■a■r■d■-■d■a■t■a■b■a■s■e■-■1■:■
■
■
```

docker-ops/03_docker_ps.txt

```
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS                    PORTS     NAMES
PORTS
NAMES

a9b96b55d9b7b7c0     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
5d/tcp, 0.0.0.0:32768->8080/tcp, :::32768->8080/tcp   hopeful_einstein

1f7c1c3b1078     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
5d/tcp, 0.0.0.0:32768->8080/tcp, :::32768->8080/tcp   hopeful_einstein

e61d4b2d2d6dbb     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
5d/tcp, 0.0.0.0:32768->8080/tcp, :::32768->8080/tcp   hopeful_einstein

1e1e6b6d6f6c     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
5d/tcp, 0.0.0.0:32768->8080/tcp, :::32768->8080/tcp   hopeful_einstein

6cd0b9b94b83fd     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
2d/tcp, 0.0.0.0:54321->8080/tcp, :::54321->8080/tcp   hopeful_einstein

c7bb3d7b4a03000     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
0.0.0.0:2707017-> 2707017/tcp, :::2707017-> 2707017/tcp,
[::]:2707017-> 2707017/tcp
hopeful_einstein

16b3e72b94b23b4     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
9d  hhoouurs  aggoo     Up  9d  hhoouurs  (hhealltthhy)
0.0.0.0:80->80/tcp, :::80->80/tcp,
[::]:80 0 0 0-> 80 0 0 0/tcp
vii sii ong ua rd- we b- 1
f3b4bec2ce3f     "/docker-entrypoi…"   2 days ago       Up 2 days                 80/tcp    hopeful_einstein
9d  hhoouurs  (hhealltthhy)
5432/tcp
vii sii ong ua rd- dat abase- 1
```

docker-ops/04_images_before.txt

```
REPOSITORY                                                        TAG       IMAGE ID       CREATED          SIZE
curriculum-pipeline-dev_app                                       latest    2b6e5d3a7f1c   2 minutes ago    2.35GB
mongo                                                             7.0       f02d7b2e1a6c   3 days ago       815MB
postgres                                                          15-alpine 9cef4d2b7e61   5 days ago       392MB
redis                                                             7-alpine  c1f4cb3d1c55   2 weeks ago      41.9MB
```

docker-ops/05_image_history.txt

```
IMAGE          CREATED        CREATED BY                                      SIZE      COMMENT
<missing>      2 months ago   /bin/sh -c #(nop)  CMD ["python3"]              0B
<missing>      2 months ago   /bin/sh -c set -ex;  wget -O get-pip.py 'ht...  9.29MB
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PYTHON_PIP_VERSION=19...  0B
<missing>      2 months ago   /bin/sh -c cd /usr/local/bin  && ln -s idle3...  32B
<missing>      2 months ago   /bin/sh -c set -ex  && wget -O python.tar.xz...  23.3MB
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PYTHON_VERSION=3.7.4      0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV GPG_KEY=0D96DF4D4110E...  0B
<missing>      2 months ago   /bin/sh -c apt-get update  && apt-get install...  4.78MB
<missing>      2 months ago   /bin/sh -c #(nop)  ENV LANG=C.UTF-8              0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PATH=/usr/local/bin:/...  0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV LD_LIBRARY_PATH=/usr/...  0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PYTHONIOENCODING=UTF-8    0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PYTHONUNBUFFERED=1        0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PYTHONDONTWRITEBYTECO...  0B
<missing>      2 months ago   /bin/sh -c #(nop)  ENV PYTHONHASHSEED=random     0B
<missing>      2 months ago   /bin/sh -c #(nop)  CMD ["/bin/bash"]             0B
<missing>      2 months ago   /bin/sh -c #(nop)  LABEL maintainer="Python ...  0B
<missing>      2 months ago   /bin/sh -c #(nop)  LABEL org.opencontainers.i
```

```
■P■Y■T■H■O■N■_■V■E■R■S■I■O■N■=■3■.■1■1■.■1■8■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■0■B■ ■ ■ ■ ■ ■ ■ ■ ■b■u■i■l■d■k■i■t■.■d■o■c■k■e■r■f■i■l■e■.■v■0■
■
■<■m■i■s■s■i■n■g■>■ ■ ■ ■ ■ ■ ■4■7■ ■h■o■u■r■s■ ■a■g■o■ ■ ■ ■E■N■V■
■G■P■G■_■K■E■Y■=■A■0■3■5■C■8■C■1■9■2■1■9■B■A■8■2■1■E■C■E■A■8■6■B■6■4■E■6■2■8■F■8■& ■
■ ■0■B■ ■ ■ ■ ■ ■ ■ ■ ■b■u■i■l■d■k■i■t■.■d■o■c■k■e■r■f■i■l■e■.■v■0■
■
■<■m■i■s■s■i■n■g■>■ ■ ■ ■ ■ ■ ■4■7■ ■h■o■u■r■s■ ■a■g■o■ ■ ■ ■R■U■N■ ■/■b■i■n■/■s■h■
■-■c■ ■s■e■t■ ■-■e■u■x■;■ ■ ■a■p■t■-■g■e■t■ ■u■p■d■a■t■e■;■ ■ ■a■& ■ ■ ■4■.■9■3■M■B■
■ ■ ■ ■b■u■i■l■d■k■i■t■.■d■o■c■k■e■r■f■i■l■e■.■v■0■
■
■<■m■i■s■s■i■n■g■>■ ■ ■ ■ ■ ■ ■4■7■ ■h■o■u■r■s■ ■a■g■o■ ■ ■ ■E■N■V■
■L■A■N■G■=■C■.■U■T■F■-■8■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■0■B■ ■ ■ ■ ■ ■ ■ ■ ■b■u■i■l■d■k■i■t■.■d■o■c■k■e■r■f■i■l■e■.■v■0■
■
■<■m■i■s■s■i■n■g■>■ ■ ■ ■ ■ ■ ■4■7■ ■h■o■u■r■s■ ■a■g■o■ ■ ■ ■E■N■V■
■P■A■T■H■=■/■u■s■r■/■l■o■c■a■l■/■b■i■n■:■/■u■s■r■/■l■o■c■a■l■/■s■b■i■n■:■/■u■s■r■& ■
■ ■0■B■ ■ ■ ■ ■ ■ ■ ■ ■b■u■i■l■d■k■i■t■.■d■o■c■k■e■r■f■i■l■e■.■v■0■
■
■<■m■i■s■s■i■n■g■>■ ■ ■ ■ ■ ■ ■2■ ■d■a■y■s■ ■a■g■o■ ■ ■ ■ ■ ■#■ ■d■e■b■i■a■n■.■s■h■
■-■-■a■r■c■h■ ■'■a■m■d■6■4■'■ ■o■u■t■/■ ■'■t■r■i■x■i■e■'■ ■'■@■1■& ■ ■ ■8■7■.■7■M■B■
■ ■ ■ ■d■e■b■u■e■r■r■e■o■t■y■p■e■ ■0■.■1■7■
■
■
```

docker-ops/08_failure_ps.txt

```
NAME                                                
```

docker-ops/09_failure_web_logs.txt

```
■■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■5■:■0■6■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■S■t■a■r■t■i■n■g■ ■g■u■n■i■c■o■r■n■ ■2■3■.■0■.■0■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■5■:■0■6■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■L■i■s■t■e■n■i■n■g■ ■a■t■:■ ■h■t■t■p■:■/■/■0■.■0■.■0■.■0■:■8■0■0■0■
■(■1■)■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■5■:■0■6■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■U■s■i■n■g■ ■w■o■r■k■e■r■:■ ■s■y■n■c■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■5■:■0■6■ ■+■0■0■0■0■]■ ■[■6■]■
■[■I■N■F■O■]■ ■B■o■o■t■i■n■g■ ■w■o■r■k■e■r■ ■w■i■t■h■ ■p■i■d■:■ ■6■
■
■w■e■b■-■1■ ■ ■|■ ■W■A■R■N■I■N■G■:■ ■A■l■l■ ■l■o■g■ ■m■e■s■s■a■g■e■s■ ■b■e■f■o■r■e■
■a■b■s■l■:■:■I■n■i■t■i■a■l■i■z■e■L■o■g■(■)■ ■i■s■ ■c■a■l■l■e■d■ ■a■r■e■
■w■r■i■t■t■e■n■ ■t■o■ ■S■T■D■E■R■R■
■
■w■e■b■-■1■ ■ ■|■ ■I■0■0■0■0■0■ ■0■0■:■0■0■:■1■7■9■4■0■3■5■7■1■7■.■5■5■5■2■7■9■ ■ ■ ■ ■
■ ■ ■6■ ■c■p■u■_■f■e■a■t■u■r■e■_■g■u■a■r■d■.■c■c■:■2■2■7■]■ ■T■h■i■s■
■T■e■n■s■o■r■F■l■o■w■ ■b■i■n■a■r■y■ ■i■s■ ■o■p■t■i■m■i■z■e■d■ ■t■o■ ■u■s■e■
■a■v■a■i■l■a■b■l■e■ ■C■P■U■ ■i■n■s■t■r■u■c■t■i■o■n■s■ ■i■n■
■p■e■r■f■o■r■m■a■n■c■e■-■c■r■i■t■i■c■a■l■ ■o■p■e■r■a■t■i■o■n■s■.■
■
■w■e■b■-■1■ ■ ■|■ ■T■o■ ■e■n■a■b■l■e■ ■t■h■e■ ■f■o■l■l■o■w■i■n■g■
■i■n■s■t■r■u■c■t■i■o■n■s■:■ ■A■V■X■2■ ■F■M■A■,■ ■i■n■ ■o■t■h■e■r■
■o■p■e■r■a■t■i■o■n■s■,■ ■r■e■b■u■i■l■d■ ■T■e■n■s■o■r■F■l■o■w■ ■w■i■t■h■ ■t■h■e■
■a■p■p■r■o■p■r■i■a■t■e■ ■c■o■m■p■i■l■e■r■ ■f■l■a■g■s■.■
■
■
```

docker-ops/12_restore_ps_healthy.txt

```
NAME   IMAGE   COMMAND   SERVICE   CREATED   STATUS   PORTS

COMMAND "docker-entrypoint.s…"

CREATED 2 seconds ago

STATUS Up 1 second (health: starting)

PORTS

volumes_postgres_data:/var/lib/postgresql/data:ro

postgres:16-alpine   "docker-entrypoint.s…"   postgres   2 seconds ago   Up 1 second (health: starting)   5432/tcp

volumes_postgres_data:/var/lib/postgresql/data:ro

volumes_postgres_data   postgres:16-alpine   "docker-entrypoint.s…"   postgres   2 seconds ago   Up 1 second (health: starting)   5432/tcp
```

docker-ops/13_restore_health.txt

```
■■
■
■S■t■a■t■u■s■C■o■d■e■ ■C■o■n■t■e■n■t■ ■ ■ 
```

docker-ops/16_mismatch_ps.txt

```
NAME  IMAGE  SERVICE
COMMAND  CREATED  STATUS
CREATED  PORTS
visionguard-web-1  visionguard-web  web
"docker-entrypoint.…"  2 hours ago  Up 2 hours
visionguard-db-1  visionguard-db  db
"docker-entrypoint.…"  2 hours ago  Up 2 hours (healthy)  5432/tcp
visionguard-worker-1  visionguard-worker  worker
"celery -A app.tasks…"  2 hours ago  Up 2 hours (healthy)  8000/tcp
```

docker-ops/17_mismatch_web_logs.txt

```
■■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■6■:■2■2■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■S■t■a■r■t■i■n■g■ ■g■u■n■i■c■o■r■n■ ■2■3■.■0■.■0■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■6■:■2■2■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■L■i■s■t■e■n■i■n■g■ ■a■t■:■ ■h■t■t■p■:■/■/■0■.■0■.■0■.■0■:■8■0■0■0■
■(■1■)■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■6■:■2■2■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■U■s■i■n■g■ ■w■o■r■k■e■r■:■ ■s■y■n■c■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■6■:■2■2■ ■+■0■0■0■0■]■ ■[■7■]■
■[■I■N■F■O■]■ ■B■o■o■t■i■n■g■ ■w■o■r■k■e■r■ ■w■i■t■h■ ■p■i■d■:■ ■7■
■
■w■e■b■-■1■ ■ ■|■ ■W■A■R■N■I■N■G■:■ ■A■l■l■ ■l■o■g■ ■m■e■s■s■a■g■e■s■ ■b■e■f■o■r■e■
■a■b■s■l■:■:■I■n■i■t■i■a■l■i■z■e■L■o■g■(■)■ ■i■s■ ■c■a■l■l■e■d■ ■a■r■e■
■w■r■i■t■t■e■n■ ■t■o■ ■S■T■D■E■R■R■
■
■w■e■b■-■1■ ■ ■|■ ■I■0■0■0■0■0■ ■0■0■:■0■0■:■1■7■9■4■0■3■5■7■8■4■.■2■2■5■4■1■0■ ■ ■ ■ ■
■ ■ ■7■ ■c■p■u■_■f■e■a■t■u■r■e■_■g■u■a■r■d■.■c■c■:■2■2■7■]■ ■T■h■i■s■
■T■e■n■s■o■r■F■l■o■w■ ■b■i■n■a■r■y■ ■i■s■ ■o■p■t■i■m■i■z■e■d■ ■t■o■ ■u■s■e■
■a■v■a■i■l■a■b■l■e■ ■C■P■U■ ■i■n■s■t■r■u■c■t■i■o■n■s■ ■i■n■
■p■e■r■f■o■r■m■a■n■c■e■-■c■r■i■t■i■c■a■l■ ■o■p■e■r■a■t■i■o■n■s■.■
■
■w■e■b■-■1■ ■ ■|■ ■T■o■ ■e■n■a■b■l■e■ ■t■h■e■ ■f■o■l■l■o■w■i■n■g■
■i■n■s■t■r■u■c■t■i■o■n■s■:■ ■A■V■X■2■ ■F■M■A■,■ ■i■n■ ■o■t■h■e■r■
■o■p■e■r■a■t■i■o■n■s■,■ ■r■e■b■u■i■l■d■ ■T■e■n■s■o■r■F■l■o■w■ ■w■i■t■h■ ■t■h■e■
■a■p■p■r■o■p■r■i■a■t■e■ ■c■o■m■p■i■l■e■r■ ■f■l■a■g■s■.■
■
■
```

docker-ops/20_final_ps_healthy.txt

```
NAME   IMAGE   COMMAND   SERVICE   CREATED   STATUS   PORTS
CREATED   STATUS
POR
vision-gua-dd-data-1   vision-gua-dd-data   "docker-entrypoint.s…"   postgres-16   2 hours ago   Up 2 hours (healthy)   5432/tcp
vision-guardian-web-1   vision-guardian-web   "guni corn -w 1 -b 0.0.0.0:8000"   web   2 hours ago   Up 2 hours (healthy)   0.0.0.0:8000->8000/tcp,
[::]:8000->8000/tcp
```

docker-ops/21_final_health.txt

```
■■
■
■S■t■a■t■u■s■C■o■d■e■ ■C■o■n■t■e■n■t■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■
■-■-■-■-■-■-■-■-■-■-■ ■-■-■-■-■-■-■-■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■
■ ■ ■ ■ ■ ■ ■ ■2■0■0■ ■{■"■m■o■d■e■l■"■:■"■v■i■s■i■o■n■g■u■a■r■d■-■c■h■a■m■p■i■o■n■"
■,■"■s■t■a■t■u■s■"■:■"■o■k■"■,■"■v■e■r■s■i■o■n■"■:■"■1■.■.■.■
■
■
■
■
■
■
```

docker-ops/23_existing_volume_wrong_password_ps.txt

```
NAMES   IMAGE   STATUS   PORTS
COMMAND
CREATED
POR TS
v i s i o n g u a r d - d a t a b a s e - 1
p o s t g r e s : 1 6 - a l p i n e   " d o c k e r - e n t r y p o i n t .
" d a t a b a s e   " 2 7   s e c o n d s   a g o   U p   2 2
s e c o n d s   ( h e a l t h y )   5 4 3 2 / t c p
```

docker-ops/24_existing_volume_wrong_password_logs.txt

```
■■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■1■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■S■t■a■r■t■i■n■g■ ■g■u■n■i■c■o■r■n■ ■2■3■.■0■.■0■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■1■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■L■i■s■t■e■n■i■n■g■ ■a■t■:■ ■h■t■t■p■:■/■/■0■.■0■.■0■.■0■:■8■0■0■0■
■(■1■)■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■1■ ■+■0■0■0■0■]■ ■[■1■]■
■[■I■N■F■O■]■ ■U■s■i■n■g■ ■w■o■r■k■e■r■:■ ■s■y■n■c■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■1■ ■+■0■0■0■0■]■ ■[■7■]■
■[■I■N■F■O■]■ ■B■o■o■t■i■n■g■ ■w■o■r■k■e■r■ ■w■i■t■h■ ■p■i■d■:■ ■7■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■3■ ■+■0■0■0■0■]■ ■[■7■]■
■[■E■R■R■O■R■]■ ■E■x■c■e■p■t■i■o■n■ ■i■n■ ■w■o■r■k■e■r■ ■p■r■o■c■e■s■s■
■
■w■e■b■-■1■ ■ ■|■ ■T■r■a■c■e■b■a■c■k■ ■(■m■o■s■t■ ■r■e■c■e■n■t■ ■c■a■l■l■
■l■a■s■t■)■:■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■u■s■r■/■l■o■c■a■l■/■l■i■b■/■p■y■t■h■o■n■3■.■1■1
■/■s■i■t■e■-■p■a■c■k■a■g■e■s■/■g■u■n■i■c■o■r■n■/■a■r■b■i■t■e■r■.■p■y■"■,■ ■l■i■n■e■
■6■0■8■,■ ■i■n■ ■s■p■a■w■n■_■w■o■r■k■e■r■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■w■o■r■k■e■r■.■i■n■i■t■_■p■r■o■c■e■s■s■(■)■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■u■s■r■/■l■o■c■a■l■/■l■i■b■/■p■y■t■h■o■n■3■.■1■1
■/■s■i■t■e■-■p■a■c■k■a■g■e■s■/■g■u■n■i■c■o■r■n■/■w■o■r■k■e■r■s■/■b■a■s■e■.■p■y■"■,■
■l■i■n■e■ ■1■3■5■,■ ■i■n■ ■i■n■i■t■_■p■r■o■c■e■s■s■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■s■e■l■f■.■l■o■a■d■_■w■s■g■i■(■)■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■u■s■r■/■l■o■c■a■l■/■l■i■b■/■p■y■t■h■o■n■3■.■1■1
■/■s■i■t■e■-■p■a■c■k■a■g■e■s■/■g■u■n■i■c■o■r■n■/■w■o■r■k■e■r■s■/■b■a■s■e■.■p■y■"■,■
■l■i■n■e■ ■1■4■7■,■ ■i■n■ ■l■o■a■d■_■w■s■g■i■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■s■e■l■f■.■w■s■g■i■ ■=■ ■s■e■l■f■.■a■p■p■.■w■s■g■i■(■)■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■^■^■^
```

```
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■a■p■p■/■a■p■p■/■d■b■.■p■y■"■,■ ■l■i■n■e■ ■1■7■,■
■i■n■ ■_■_■i■n■i■t■_■_■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■s■e■l■f■.■i■n■i■t■i■a■l■i■z■e■(■)■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■a■p■p■/■a■p■p■/■d■b■.■p■y■"■,■ ■l■i■n■e■ ■3■2■,■
■i■n■ ■i■n■i■t■i■a■l■i■z■e■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■w■i■t■h■ ■s■e■l■f■.■c■o■n■n■e■c■t■(■)■ ■a■s■
■c■o■n■n■e■c■t■i■o■n■:■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■a■p■p■/■a■p■p■/■d■b■.■p■y■"■,■ ■l■i■n■e■ ■2■3■,■
■i■n■ ■c■o■n■n■e■c■t■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■c■o■n■n■e■c■t■i■o■n■ ■=■
■p■s■y■c■o■p■g■.■c■o■n■n■e■c■t■(■s■e■l■f■.■d■a■t■a■b■a■s■e■_■u■r■l■)■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■^■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■F■i■l■e■ ■"■/■u■s■r■/■l■o■c■a■l■/■l■i■b■/■p■y■t■h■o■n■3■.■1■1
■/■s■i■t■e■-■p■a■c■k■a■g■e■s■/■p■s■y■c■o■p■g■/■c■o■n■n■e■c■t■i■o■n■.■p■y■"■,■
■l■i■n■e■ ■1■1■9■,■ ■i■n■ ■c■o■n■n■e■c■t■
■
■w■e■b■-■1■ ■ ■|■ ■ ■ ■ ■ ■r■a■i■s■e■
■l■a■s■t■_■e■x■.■w■i■t■h■_■t■r■a■c■e■b■a■c■k■(■N■o■n■e■)■
■
■w■e■b■-■1■ ■ ■|■ ■p■s■y■c■o■p■g■.■O■p■e■r■a■t■i■o■n■a■l■E■r■r■o■r■:■
■c■o■n■n■e■c■t■i■o■n■ ■f■a■i■l■e■d■:■ ■c■o■n■n■e■c■t■i■o■n■ ■t■o■ ■s■e■r■v■e■r■ ■a■t■
■"■1■7■2■.■1■8■.■0■.■2■"■,■ ■p■o■r■t■ ■5■4■3■2■ ■f■a■i■l■e■d■:■ ■F■A■T■A■L■:■ ■
■p■a■s■s■w■o■r■d■ ■a■u■t■h■e■n■t■i■c■a■t■i■o■n■ ■f■a■i■l■e■d■ ■f■o■r■ ■u■s■e■r■
■"■v■i■s■i■o■n■g■u■a■r■d■"■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■3■ ■+■0■0■0■0■]■ ■[■7■]■
■[■I■N■F■O■]■ ■W■o■r■k■e■r■ ■e■x■i■t■i■n■g■ ■(■p■i■d■:■ ■7■)■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■3■ ■+■0■0■0■0■]■ ■[■1■]■
■[■E■R■R■O■R■]■ ■W■o■r■k■e■r■ ■(■p■i■d■:■7■)■ ■e■x■i■t■e■d■ ■w■i■t■h■ ■c■o■d■e■ ■3■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■3■ ■+■0■0■0■0■]■ ■[■1■]■
■[■E■R■R■O■R■]■ ■S■h■u■t■t■i■n■g■ ■d■o■w■n■:■ ■M■a■s■t■e■r■
■
■w■e■b■-■1■ ■ ■|■ ■[■2■0■2■6■-■1■1■-■0■7■ ■0■7■:■1■7■:■3■3■ ■+■0■0■0■0■]■ ■[■1■]■
■[■E■R■R■O■R■]■ ■R■e■a■s■o■n■:■ ■W■o■r■k■e■r■ ■f■a■i■l■e■d■ ■t■o■ ■b■o■o■t■.■
■
■
```

docker-ops/26_fixed_password_ps.txt

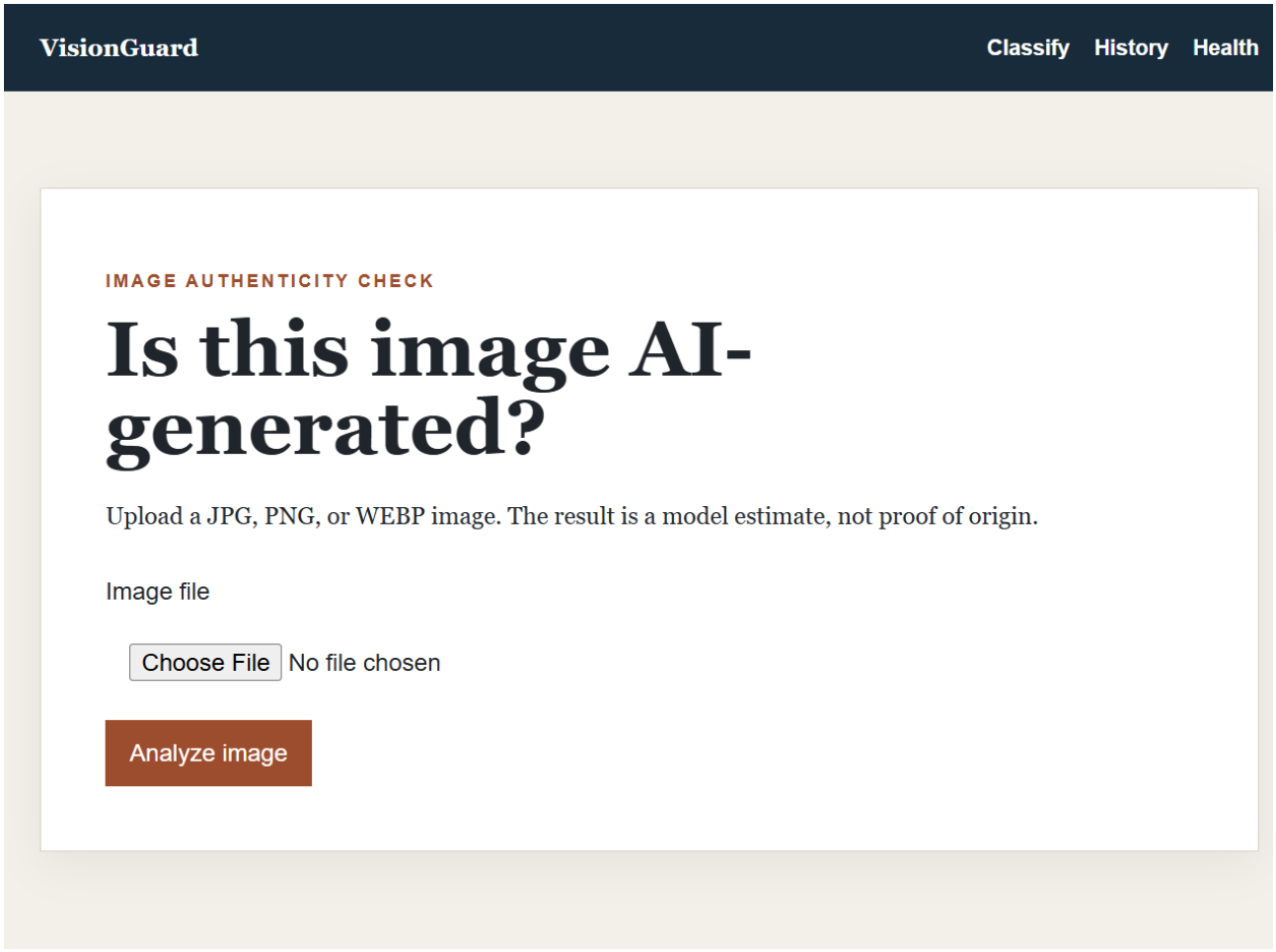
```
NAMES   IMAGE   COMMAND   SERVICE   CREATED   STATUS   PORTS
CREATED   STATUS
POSTS
vision-gunicorn-django-db-1   vision-gunicorn-django   "docker-entrypoint.s…"   db   2 days ago   Up 2 days
vision-gunicorn-django-1   vision-gunicorn-django   "/entrypoint.sh /sta…"   django   2 days ago   Up 2 days
vision-gunicorn-django-web-1   vision-gunicorn-django   "/entrypoint.sh /sta…"   web   2 days ago   Up 2 days   0.0.0.0:8000->8000/tcp,
[::]:8000->8000/tcp
```

live-validation/01_health.txt

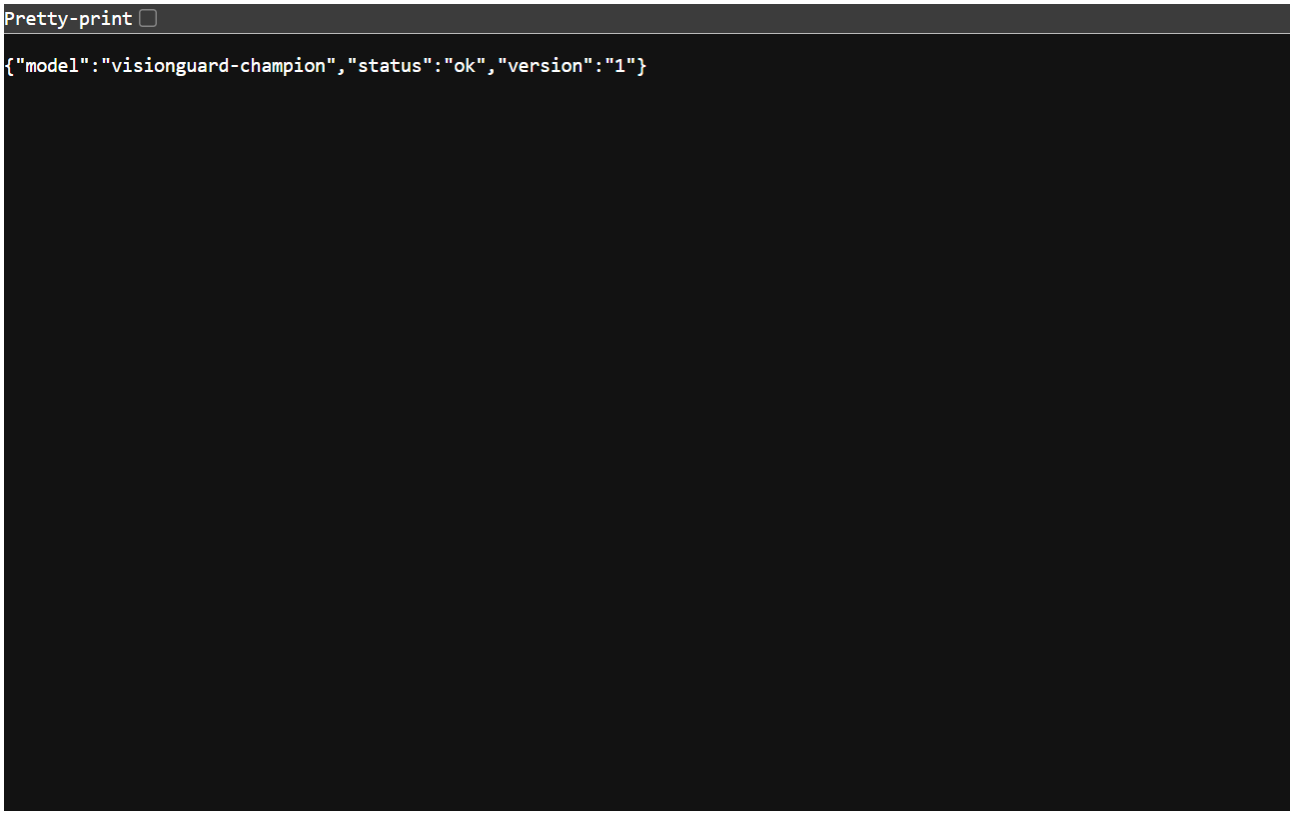
live-validation/01_health_error.txt

```
■■I■n■v■o■k■e■-■W■e■b■R■e■q■u■e■s■t■ ■:■ ■T■h■e■ ■r■e■m■o■t■e■ ■s■e■r■v■e■r■
■r■e■t■u■r■n■e■d■ ■a■n■ ■e■r■r■o■r■:■ ■(■5■0■3■)■ ■S■e■r■v■e■r■
■U■n■a■v■a■i■l■a■b■l■e■.■
■
■A■t■ ■l■i■n■e■:■2■ ■c■h■a■r■:■1■4■4■
■
■+■ ■.■.■.■ ■N■u■l■l■;■ ■t■r■y■ ■{■ ■I■n■v■o■k■e■-■W■e■b■R■e■q■u■e■s■t■
■-■U■s■e■B■a■s■i■c■P■a■r■s■i■n■g■ ■'■h■t■t■p■s■:■/■/■2■3■l■-■0■5■8■8■-■a■2■-■o■
■.■.■.■
■
■+■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■~■~■~
```

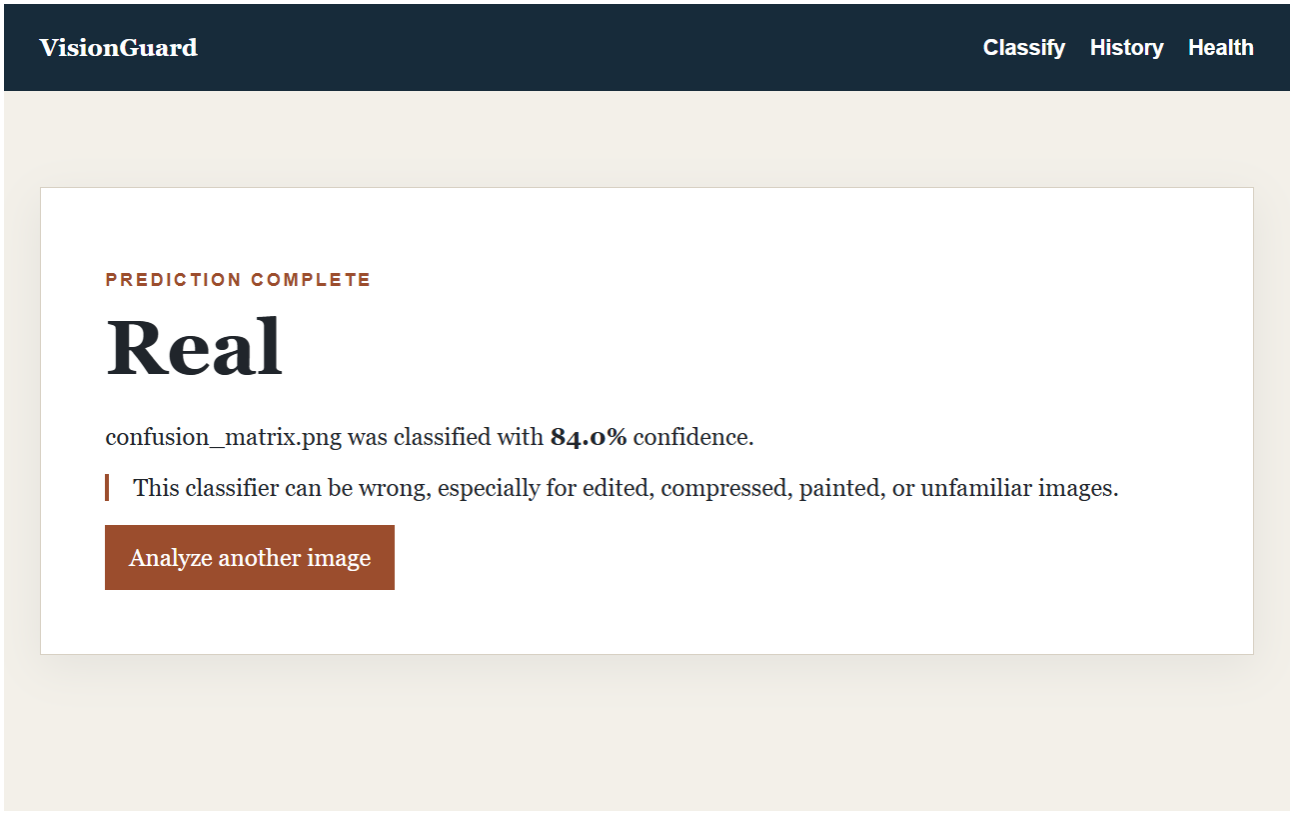
Live Deployment Evidence



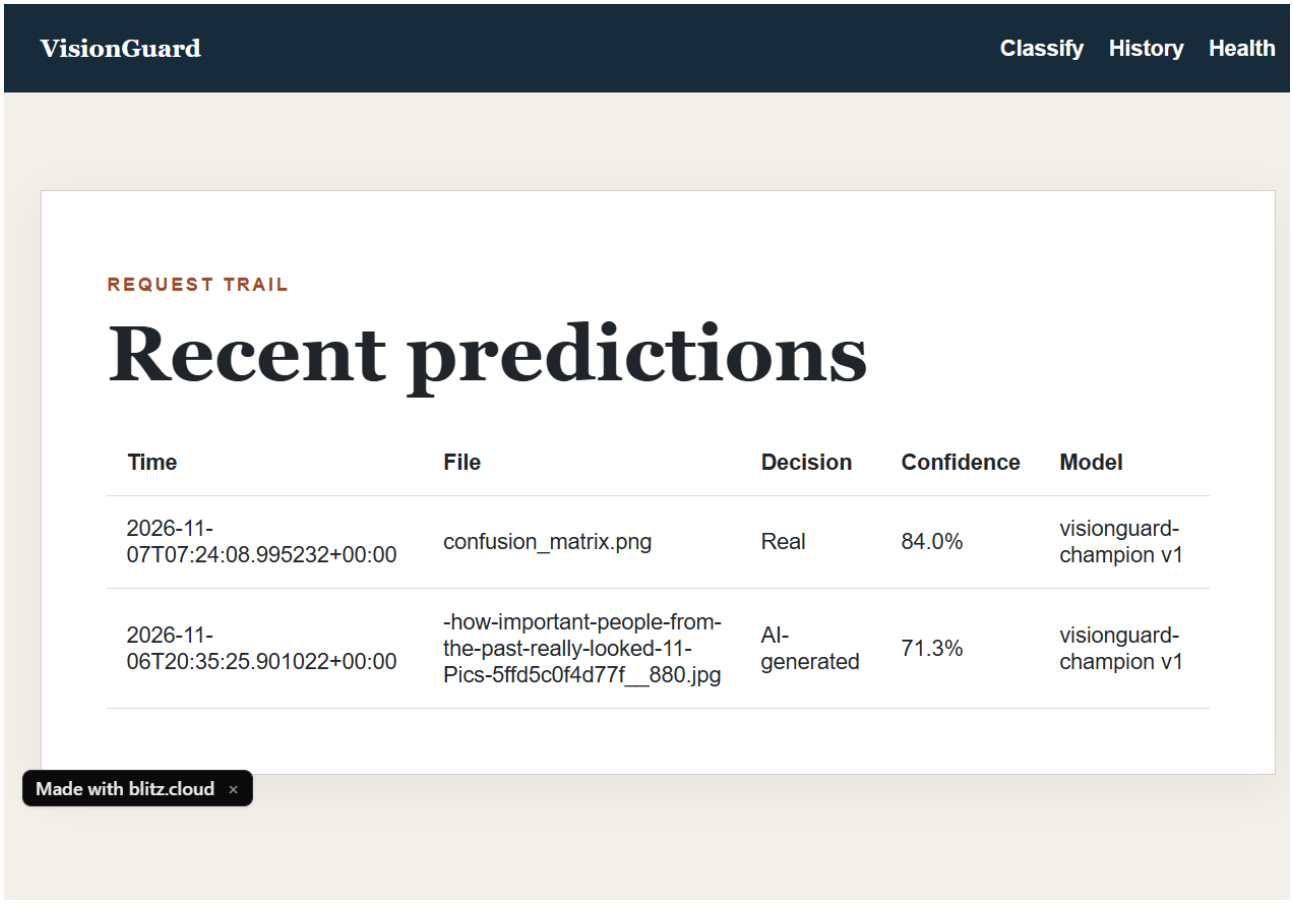
31_live_public_ui_awake.png — Live public VisionGuard upload interface.



32_live_health_awake.png — Live /health response with champion model, version 1, and ok status.



33_live_prediction.png — Live /predict result from a real image upload.



34_live_history.png — Live /history page with persisted prediction records.

Live API Request Evidence

Equivalent multipart request executed with curl because native Postman window capture was unavailable.

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>VisionGuard</title>
<link rel="stylesheet" href="/static/styles.css">
</head>
<body>
<header class="site-header">
<a href="/">VisionGuard</a>
<nav><a href="/">Classify</a><a href="/history">History</a><a href="/health">Health</a></nav>
</header>
<main class="container">



<section class="card result">
<p class="eyebrow">Prediction complete</p>
<h1>Real</h1>
<p>confusion_matrix.png was classified with <strong>84.0%</strong> confidence.</p>
<p class="caveat">This classifier can be wrong, especially for edited, compressed, painted, or
unfamiliar images.</p>
<a class="button" href="/">Analyze another image</a>
</section>

</main>
</body>
</html>
```